

# A FINITELY PRESENTED GROUP THAT IS NOT MF

## ANONYMOUS

ABSTRACT. A countable discrete group is MF if it embeds in the unitary group of a quotient  $\prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C})$ ; equivalently, if it admits unitary matrix models that are asymptotically multiplicative in operator norm and separate every nontrivial element from the identity. We construct a finitely presented group  $E$ , on eight generators and forty-one relators, that is not MF. This refutes the conjecture that every countable group is MF.

Let  $\Gamma \leq H$  have property (T), let  $t\Gamma t^{-1} \subseteq \Gamma$ , let  $c$  centralize  $\Gamma$ , and put  $d = tct^{-1}$  and  $u = [d, a]$  for  $a \in \Gamma$ . We prove that in every operator-norm matrix model of  $H$  conjugation by  $t$  preserves the Hilbert–Schmidt asymptotic commutant of  $\Gamma$ ; hence  $\|u - 1\|_2 \rightarrow 0$ . In  $E$  the element  $w = u^2$  is central of order two, and a Clifford-group quotient of  $E$  sends it to  $-1$ . So  $w \neq 1$  while every matrix model sends  $w$  to 1. The construction takes as input any finitely presented property-(T) group with a proper injective self-embedding;  $E$  is the instance built on  $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ .

The quotient  $E/\langle w \rangle$  is again not MF, so the MF radical of  $E$  properly contains  $\{1, w\}$ . The reduced group  $C^*$ -algebra  $C_{\mathrm{red}}^*(E)$  is separable and stably finite but not MF, and the Clifford quotient of  $E$  that proves  $w \neq 1$  is a finitely generated sofic — hence hyperlinear — group that is not MF.

## 1 INTRODUCTION

Can every countable discrete group be realized by matrices, if multiplication is only required to hold up to operator-norm errors that vanish along a sequence of finite dimensions? Shulman calls the positive assertion the *MF conjecture* [Sh], and no group was known to fail it [Sc, Sh]. We prove that the answer is no: there is an explicit finitely presented group, on eight generators and forty-one relators, that is not MF. Residually finite groups satisfy the conjecture through their finite quotients, as do LEF groups [CDE] and, by quasidiagonality [TWW], all amenable groups. Recent work establishes it for many amalgamated free products [Sc, Sh, GKEMP]. The corresponding  $C^*$ -algebra question is older. Blackadar and Kirchberg, who introduced MF algebras, remarked that no stably finite separable  $C^*$ -algebra was known not to be MF [BK, CDE]. Their natural candidate was  $C_{\mathrm{red}}^*(F_2)$ , which Haagerup and Thorbjørnsen later proved to be MF [HT]. The negative

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solution of the Connes embedding problem subsequently supplied abstract stably finite non-MF algebras [JNVWY, FGH]. No countable group was known whose reduced or maximal group  $C^*$ -algebra fails to be MF [Sh].

These questions belong to a program of finitary approximation. In his 2018 ICM address, Thom asked for which Schatten  $p$ -norms there exist non-approximable groups [Th, Question 3.11]. The case  $p = 2$  was settled by De Chiffre–Glebsky–Lubotzky–Thom [DGLT], the cases  $1 < p < \infty$  by Lubotzky–Oppenheim [LuO], and the case  $p = 1$  by Bachner–Dogon–Lubotzky [BDL]. That left the Hamming metric (soficity), the Hilbert–Schmidt norm (hyperlinearity), and the operator norm (MF). The Hamming case was disproved in 2026: the first nonsofic group was constructed by OpenAI [OAI], Fournier–Facio then gave a torsion-free example [FFF], and Kun–Thom developed further wreath-product examples [KT26]. One-sided conjugation of a property-(T) subgroup into itself drives those arguments and also starts this one. Soficity concerns permutations in the Hamming metric and MF concerns unitaries in operator norm, so a different rigidity argument is needed; we obtain it from a one-sided transport theorem for the Kazhdan fixed spaces of the adjoint actions. Of the three problems, only hyperlinearity remains open. In the language of approximate representations, a countable discrete group  $G$  is MF if, for every finite subset  $F \subseteq G$  and every  $\varepsilon > 0$ , there is a unitary matrix model  $V : G \rightarrow \mathcal{U}(r)$  that is multiplicative within  $\varepsilon$  in operator norm on  $F$  and separates distinct elements of  $F$  by operator-norm distance at least 1. There is an equivalent corona formulation, in which the notion was introduced by Carrión–Dadarlat–Eckhardt and is used by Shulman and Bachner–Dogon–Lubotzky [CDE, Sh, BDL]. Let  $(d_n)_{n \in \mathbb{N}}$  be a strictly increasing sequence of positive integers and let

$$(1) \quad \mathcal{Q} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C})$$

be the associated matrix quotient: bounded sequences of matrices modulo sequences converging to 0 in operator norm. We call any such corona algebra a *norm matrix corona*. Following Blackadar–Kirchberg [BK], a separable  $C^*$ -algebra is MF if it embeds into some algebra of the form (1); the group  $G$  is MF if there is an injective homomorphism  $G \rightarrow \mathcal{U}(\mathcal{Q})$ . The two formulations are equivalent (Proposition 1.1). We shall use the equivalent coordinate model

$$(2) \quad \mathcal{U}_{\text{cor}}((d_n)) := \prod_n \mathcal{U}(M_{d_n}(\mathbb{C})) / \{(u_n) : \|u_n - 1\| \rightarrow 0\}.$$

The denominator is normal, and polar correction gives a canonical isomorphism

$$(3) \quad \kappa_{(d_n)} : \mathcal{U}_{\text{cor}}((d_n)) \xrightarrow{\cong} \mathcal{U}(\mathcal{Q}),$$

proved in Lemma 2.2. Thus one may equivalently embed  $G$  into  $\mathcal{U}_{\text{cor}}((d_n))$ .

**Proposition 1.1** (equivalent MF formulations). *For every countable group, the  $C^*$ -corona definition above is equivalent to the unitary-sequence definition.*

*The latter is equivalent to operator-norm local models with separation constant 1, and allowing arbitrary positive dimensions is equivalent to requiring the dimensions to be strictly increasing.*

The proof is deferred, together with the precise formulation of the operator-norm local models, to Section S1.

Hereafter “MF” means this property: the operator-norm notion of Carrión–Dadarlat–Eckhardt, in the form used by Shulman. Stronger conventions exist. In [GKEMP] the approximating maps must also recover the canonical trace and the left-regular norm on every group-ring element. The purely matricial field (PMF) convention of [MdlS, Definition 1.2] requires finite-dimensional representations whose group-ring norms converge to the regular norms.

Throughout,  $[g, h] = ghg^{-1}h^{-1}$  for group elements, while for matrices  $[x, y] = xy - yx$ ; the arguments always make the reading unambiguous.

### The result

**Theorem A** (explicit Kazhdan–Clifford group). *Let  $E$  be the group on the eight generators  $v_1, v_2, v_3, x, y, z, t, c$  with the forty-one relators displayed in Definition S2.1. Put*

$$w = [tct^{-1}, v_1(tct^{-1})v_1^{-1}].$$

*The defining relations make  $w$  central; together with  $c^2 = 1$ , centrality implies  $w^2 = 1$ . Then:*

- (1)  *$w \neq 1$  in  $E$ , so  $w$  is a central involution;*
- (2) *for every sequence  $(d_n)$  of positive integers and every group homomorphism  $\Theta: E \rightarrow \mathcal{U}_{\text{cor}}((d_n))$  into the corresponding unitary norm corona (2), one has  $\Theta(w) = 1$ .*

*In particular  $E$  is not MF, and neither  $C_{\max}^*(E)$  nor  $C_{\text{red}}^*(E)$  is an MF  $C^*$ -algebra.*

The group is written out in Definition S2.1; Section 6 proves the theorem.

**The mechanism** Write  $\sim_2$  for convergence in normalized Hilbert–Schmidt norm.

Let  $\Gamma$  be a property-(T) group with a homomorphism  $\iota: \Gamma \rightarrow H$ , and let  $t, c \in H$  satisfy

$$t\iota(\Gamma)t^{-1} \subseteq \iota(\Gamma), \quad [c, \iota(\Gamma)] = 1.$$

An operator-norm almost representation of  $H$  acts by conjugation on the Hilbert–Schmidt space of each matrix block, and operator-norm control keeps this adjoint action approximately multiplicative. Let  $P$  be the projection onto the vectors fixed by the adjoint action of  $\iota(\Gamma)$  and let  $V$  be the adjoint unitary of  $t$ ; the spectral gap of a Kazhdan pair places  $P$  in the norm ultraproduct  $B_\omega$  of the blocks. The inclusion  $t\iota(\Gamma)t^{-1} \subseteq \iota(\Gamma)$  gives  $P \leq VPV^*$ . The projections  $P$  and  $VPV^*$  are Murray–von Neumann equivalent, and  $B_\omega$  contains no proper isometry, so  $P = VPV^*$ . Hence conjugation by  $t$  preserves the Hilbert–Schmidt asymptotic commutant of  $\iota(\Gamma)$  (Theorem 3.1). Section S5 proves the same statement at each fixed matrix dimension, with

an almost-fixed spectral subspace in place of  $P$  and an equal-rank reversal estimate in place of finiteness.

Put  $d = tct^{-1}$  and, for  $a \in \Gamma$ ,

$$u = [d, \iota(a)].$$

Since  $c$  commutes with  $\iota(\Gamma)$ , so does  $d$  asymptotically, and therefore

$$u \sim_2 1$$

in every operator-norm matrix model of  $H$ .

Suppose now that  $\varepsilon = u^2$  is central with  $\varepsilon^2 = 1$ , and let  $\Theta$  be a corona representation with  $\Theta(\varepsilon) \neq 1$ . Then  $\Theta(\varepsilon)$  is a self-adjoint unitary commuting with  $\Theta(H)$ , so

$$q = \frac{1}{2}(1 - \Theta(\varepsilon))$$

is a nonzero projection commuting with  $\Theta(H)$ . Lifting  $\Theta$  to coordinates and compressing each block to  $q$  gives a second operator-norm matrix model of  $H$ , on blocks each carrying their own normalized trace, in which  $\varepsilon = -1$ . Transport applies to that model and gives  $u^2 \sim_2 1$ , while  $\|-1 - 1\|_2 = 2$ . Hence  $\Theta(\varepsilon) = 1$  for every corona representation  $\Theta$ , and if  $\varepsilon \neq 1$  then  $H$  is not MF (Theorem 4.1).

What remains is  $\varepsilon \neq 1$ , a question about the group alone. Adjoin  $t$  and  $c$  to a finitely presented Kazhdan group  $\Gamma$  along a proper injective endomorphism  $\alpha$ , and let  $a$  lie outside  $\alpha(\Gamma)$ . In the mapping telescope of  $\alpha$  the cosets  $t\Gamma$  and  $at\Gamma$  are distinct, because  $a \notin \alpha(\Gamma)$ . Send those two cosets to anticommuting Clifford generators: then  $u$  is a product of two anticommuting involutions, so  $u^2 = -1$  in that quotient and  $w = u^2 \neq 1$  (Theorem 5.1).

Taking

$$\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z}), \quad \alpha(v, A) = (2v, A), \quad a = (e_1, I)$$

gives the group  $E$  of Theorem A.

The quotient  $E/\langle w \rangle$  is again not MF: the image of  $u$  is nontrivial there and lies in the kernel of every corona representation, so  $\{1, w\}$  is a proper subgroup of the MF radical of  $E$  (Theorem B). The reduced group  $C^*$ -algebra  $C_{\mathrm{red}}^*(E)$  is separable and stably finite but not MF (Theorem C). The Clifford quotient  $W$  used to prove  $w \neq 1$  is finitely generated, sofic, and not MF, so neither soficity nor hyperlinearity implies the MF property (Theorem D); whether  $E$  itself is sofic is open (Section S12).

A Lean 4 formalization of the results below accompanies the paper.

**Relation to prior work** The closest analogue is the recent work on nonsocfic groups. One-sided conjugation of a property-(T) subgroup, with propagation from its centralizer, appears in [OAI, KT26], building on rigidity results of Kun and Kun–Thom [Ku16, KT19]; Alekseev–Thom prove a related centralizer rigidity for sofic embeddings [AT]. Those results concern Hamming or tracial approximations. Here the adjoint actions stay on the Hilbert–Schmidt spaces of the matrix blocks, and  $C^*$ -finiteness of the norm ultraproduct

forbids a proper one-sided compression of the Kazhdan fixed space; the quantitative form is the equal-rank reversal estimate of Section S5.

Slofstra’s hyperlinear-profile construction uses a Clifford group, a distinguished central involution, a shift, and an HNN doubling map [Slo18, Section 2 and Remark 3.3], and his quotients by the elements invisible to exact or approximate finite-dimensional representations precede the MF radical used below [Slo17, Definitions 2.5–2.7]. Negative-eigenspace compression appears in Slofstra–Vidick [SV, proof of Proposition 2.7], and Fritz’s marked  $BS(2, 3)$  example is the context for the cyclic comparison of Section S9 [Fri, SV]. Bachner–Dogon–Lubotzky develop the operator-norm rounding, compression, and polar-correction procedures used in the finite-normal corner reduction [BDL, Lemmas 2.2–2.3, Proposition 2.4, and the proof of Proposition 1.5]; in the proof of their Proposition 1.6 they treat the finite-central case by a character-isotypic corner and reduce the general finite-normal case to it through the finite-index centralizer. Theorem S7.3 below combines that coordinate corner with one-sided Kazhdan conjugation, and its hypotheses constrain only the subgroup, not the quotient. Central extensions and rigidity of approximate representations also drive the Frobenius obstruction of De Chiffre–Glebsky–Lubotzky–Thom [DGLT]; finite-target approximation and extension results for weak soficity appear in [GR, Gl].

The passage from operator norm to the Hilbert–Schmidt conjugation representation belongs to the  $\pi \otimes \bar{\pi}$  framework of Bekka and Bekka–Valette [Be, BV], and Dadarlat uses normalized finite-rank projections as almost invariant Hilbert–Schmidt vectors in an operator-norm approximation argument [Dad, Lemma 3.18 and Proposition 3.19]. The group-MF framework is that of Carrión–Dadarlat–Eckhardt, who also give, through Abels’ group, an amenable MF group that is not residually finite [CDE, Section 2.14 and Corollary 2.18].

The group property is weaker than the algebra property: it asks only for a faithful canonical image in some MF quotient of  $C^*_{\max}(E)$ , so failure of MF for a group  $C^*$ -algebra leaves open whether the group embeds in a corona. For  $E$  both fail, and for the same reason.

A supplementary part follows Section 7: it verifies the explicit presentation, develops the mechanism in its general form, and draws the remaining consequences.

## 2 NORM MATRIX CORONAS

All groups are discrete and countable. Property (T) is the textbook complex-unitary property. A  $C^*$ -homomorphism is unital only when this is stated. In particular, an MF embedding may be nonunital, as is customary for embedding a unital algebra as a  $C^*$ -subalgebra. If  $e: A \rightarrow B$  is an injective, possibly nonunital  $*$ -homomorphism between unital  $C^*$ -algebras, then  $u \mapsto e(u) + (1 - e(1))$  is an injective homomorphism  $\mathcal{U}(A) \rightarrow \mathcal{U}(B)$ ; this complement correction is used below. For a unital  $C^*$ -algebra  $A$  we write  $\mathcal{U}(A)$

for its unitary group. On  $M_r(\mathbb{C})$  we use the normalized trace  $\text{tr}_r$ , the operator norm  $\|\cdot\|$ , and the normalized Hilbert–Schmidt norm  $\|x\|_2 = \text{tr}_r(x^*x)^{1/2}$ . For coordinate sequences we write  $a \sim_2 b$  when  $\|a_n - b_n\|_2 \rightarrow 0$ . We use without comment:  $\|x\|_2 \leq \|x\|$ ;  $|\text{tr}_r(x)| \leq \|x\|$ ; and  $\|uxv\|_2 = \|x\|_2$  for unitaries  $u, v$ . For a nonempty finite set  $Y$ , write  $M_Y(\mathbb{C}) = \mathbb{C}^{Y \times Y}$  for the corresponding indexed matrix algebra.

Given a sequence  $(d_n)_{n \in \mathbb{N}}$  of positive integers, we write

$$\mathcal{Q}((d_n)_{n \in \mathbb{N}}) = \prod_{n \in \mathbb{N}}^{\ell^\infty} M_{d_n}(\mathbb{C}) / \bigoplus_{n \in \mathbb{N}}^{c_0} M_{d_n}(\mathbb{C}),$$

the quotient of the bounded  $C^*$ -product by the ideal of sequences with  $\|x_n\| \rightarrow 0$ . We call any such algebra a *norm matrix corona*. We also write

$$\mathcal{U}_{\text{cor}}((d_n)_{n \in \mathbb{N}}) = \prod_{n \in \mathbb{N}} \mathcal{U}(M_{d_n}(\mathbb{C})) / \{(u_n) : \|u_n - 1\| \rightarrow 0\}.$$

Write  $q: \prod_{n \in \mathbb{N}}^{\ell^\infty} M_{d_n}(\mathbb{C}) \rightarrow \mathcal{Q}$  for the quotient map of a norm matrix corona.

The following lifting facts are standard perturbation theory; see Loring [Lo] for a systematic treatment.

**Lemma 2.1** (lifting). *Every unitary  $u \in \mathcal{Q}$  lifts to a sequence of unitaries.*

*Proof.* Lift  $u$  to a bounded sequence  $(x_n)$ ; unitarity of  $q((x_n))$  gives  $\|x_n^*x_n - 1\| \rightarrow 0$ . Once  $\|x_n^*x_n - 1\| \leq \frac{1}{2}$ , the matrix  $x_n$  is invertible and the polar correction  $u_n = x_n(x_n^*x_n)^{-1/2}$  is unitary with  $\|u_n - x_n\| \leq 2\|x_n\|\|x_n^*x_n - 1\| \rightarrow 0$  by continuous functional calculus; set  $u_n = 1$  at the finitely many remaining indices.  $\square$

**Lemma 2.2** (unitaries in the quotient). *For every sequence  $(d_n)$ , the map*

$$\kappa_{(d_n)}: \mathcal{U}_{\text{cor}}((d_n)) \longrightarrow \mathcal{U}(\mathcal{Q}), \quad [(u_n)] \longmapsto q((u_n)),$$

*is a canonical group isomorphism.*

*Proof.* The kernel consists exactly of the unitary sequences with  $\|u_n - 1\| \rightarrow 0$ , giving injectivity; surjectivity is Lemma 2.1.  $\square$

A *corona representation* of a group  $H$  is a homomorphism  $H \rightarrow \mathcal{U}(\mathcal{Q})$ , and it *detects*  $g \in H$  — separates  $g$  from the identity — when it does not send  $g$  to 1. An *operator-norm asymptotic representation* of  $H$  is a sequence of maps  $U_n: H \rightarrow \mathcal{U}(d_n)$  such that  $\|U_n(g)U_n(h) - U_n(gh)\| \rightarrow 0$  for all  $g, h \in H$ . Lemma 2.2 canonically identifies a corona representation with a homomorphism into  $\mathcal{U}_{\text{cor}}((d_n))$ , which we call its coordinate model. A countable group is *MF* if it admits an injective corona representation for a strictly increasing dimension sequence; by the dimension-amplification equivalence, allowing an arbitrary positive dimension sequence gives the same class of groups. Taking  $g = h = 1$  gives  $\|U_n(1) - 1\| \rightarrow 0$  automatically, since  $U_n(1)$  is unitary. Coordinate unitary lifts of a corona representation of a countable group form exactly such a sequence.

Section S13 constructs the maximal group  $C^*$ -algebra  $C_{\max}^*(H)$  and proves its universal property.

### 3 ONE-SIDED KAZHDAN TRANSPORT

Three finite statements of one principle:

$$\begin{array}{ccc} X \subseteq Y, |X| = |Y| & V \subseteq W, \dim V = \dim W & \begin{array}{c} p \leq q, p \sim q \\ \text{in a finite } C^*\text{-algebra} \end{array} \\ \Downarrow & \Downarrow & \Downarrow \\ X = Y & V = W & p = q. \end{array}$$

In the right-hand column,  $p \sim q$  is Murray–von Neumann equivalence and *finite* is meant in the  $C^*$ -sense: no proper isometries. Theorem S8.1 uses the middle implication: conjugation carries the commutant of the base into itself, and equal finite dimension forces equality. The proof below uses the right-hand one: property (T) supplies a Kazhdan projection  $P$  in a norm ultraproduct, the one-sided conjugation gives  $P \leq VPV^*$ , and finiteness forces equality. Section S5 runs the same argument quantitatively at each single coordinate, with an almost-fixed spectral subspace in place of the Kazhdan projection and an equal-rank reversal estimate in place of finiteness.

**Theorem 3.1** (Kazhdan transport of Hilbert–Schmidt asymptotic commutants). *Let  $\Gamma$  and  $H$  be groups, let  $\Gamma$  have property (T), let  $\iota: \Gamma \rightarrow H$  be a homomorphism, let  $s \in H$ , and let  $(d_n)$  be a sequence of positive integers. Suppose*

$$s\iota(\Gamma)s^{-1} \subseteq \iota(\Gamma).$$

*Let  $U_n: H \rightarrow \mathcal{U}(d_n)$  be an operator-norm asymptotic representation. If  $(x_n)$  is uniformly operator-norm bounded and asymptotically centralizes  $\iota(\Gamma)$  in normalized Hilbert–Schmidt norm,*

$$\|x_n U_n(\iota(\gamma)) - U_n(\iota(\gamma)) x_n\|_2 \longrightarrow 0 \quad (\gamma \in \Gamma),$$

*then the conjugated sequence does as well:*

$$\|[U_n(s)x_n U_n(s)^*, U_n(\iota(\gamma))]\|_2 \longrightarrow 0 \quad (\gamma \in \Gamma).$$

*Proof.* Suppose the conclusion fails: there are  $\gamma_0 \in \Gamma$ ,  $\delta > 0$ , and an infinite set  $I \subseteq \mathbb{N}$  with  $\|[U_n(s)x_n U_n(s)^*, U_n(\iota(\gamma_0))]\|_2 \geq \delta$  for  $n \in I$ . Fix a free ultrafilter  $\omega$  on  $\mathbb{N}$  with  $I \in \omega$ .

*The adjoint model.* Regard  $M_{d_n}(\mathbb{C})$  as the Hilbert space  $K_n = L^2(M_{d_n}(\mathbb{C}), \text{tr}_{d_n})$  and let each group element act on it by conjugation,  $\text{Ad } U_n(g) \xi = U_n(g) \xi U_n(g)^*$ . Operator-norm almost multiplicativity of  $U_n$  makes  $g \mapsto \text{Ad } U_n(g)$  an operator-norm asymptotic representation on these Hilbert spaces; this is the finite-stage form of the  $\pi \otimes \bar{\pi}$  construction.

*The ultraproduct.* Let  $K_\omega$  be the Hilbert-space ultraproduct of the  $K_n$  along  $\omega$  and let  $B_\omega = \prod_\omega B(K_n)$  be the norm ultraproduct, acting on  $K_\omega$  by  $[A_n]_\omega [\xi_n]_\omega = [A_n \xi_n]_\omega$ ; write  $[\cdot]_\omega$  for classes in either. This action is faithful: if  $A = [A_n]_\omega \neq 0$  then  $\lim_\omega \|A_n\| = \delta > 0$ , and unit vectors  $\xi_n$  with  $\|A_n \xi_n\| \geq \|A_n\| - \frac{1}{n}$  give  $\|A[\xi_n]_\omega\| = \lim_\omega \|A_n \xi_n\| = \delta$ . An inclusion of ranges in  $K_\omega$  may therefore be read back as an inequality of projections in  $B_\omega$ : if



$\text{ran } P \subseteq \text{ran } Q$  for projections  $P, Q \in B_\omega$ , then  $QP$  and  $P$  act identically on  $K_\omega$ , so  $QP = P$ , that is,  $P \leq Q$ . Asymptotic multiplicativity makes

$$\pi: H \longrightarrow \mathcal{U}(B_\omega), \quad \pi(g) = [\text{Ad } U_n(g)]_\omega,$$

a homomorphism. The algebra  $B_\omega$  is *finite*. Indeed, suppose  $\sigma = [\sigma_n]_\omega$  satisfies  $\sigma^* \sigma = 1$ . Then  $\|\sigma_n^* \sigma_n - 1\| \rightarrow 0$  along  $\omega$ , and each  $\sigma_n$  acts on a finite-dimensional space. Polar correction therefore replaces  $\sigma_n$  by a unitary  $w_n$  with  $\|\sigma_n - w_n\| \rightarrow 0$ . Hence  $\sigma = [w_n]_\omega$  is unitary.

*The Kazhdan projection.* Property (T) provides a finite symmetric generating set  $S \subseteq \Gamma$  with  $1 \in S$  and a Kazhdan constant  $\kappa \in (0, 1]$  for the pair  $(S, \kappa)$ . Let  $h$  be the Hermitian part of  $\frac{1}{|S|} \sum_{s' \in S} \pi(\iota(s'))$  and let  $\text{Fix} \subseteq K_\omega$  be the subspace of vectors fixed by every  $\pi(\iota(\gamma))$ . Both  $\text{Fix}$  and its orthocomplement are invariant under each unitary  $\pi(\iota(s'))$ , hence under  $h$ . On  $\text{Fix}$ ,  $h = 1$ . The restriction of  $\pi \circ \iota$  to  $\text{Fix}^\perp$  has no nonzero invariant vector, so a unit vector  $\xi \in \text{Fix}^\perp$  satisfies  $\|\pi(\iota(s'))\xi - \xi\| \geq \kappa$  for some generator. For that generator  $\text{Re}\langle \pi(\iota(s'))\xi, \xi \rangle \leq 1 - \kappa^2/2$ , while every other term is at most 1. Averaging gives  $\langle h\xi, \xi \rangle \leq 1 - \kappa^2/(2|S|)$ . Consequently

$$\text{sp}(h) \subseteq \left[-1, 1 - \frac{\kappa^2}{2|S|}\right] \cup \{1\},$$

and the spectral projection  $P$  of  $h$  at the isolated point 1 belongs to  $B_\omega$  by continuous functional calculus and projects onto  $\text{Fix}$ .

*One-sided compression and finiteness.* Let  $V = \pi(s)$  and  $Q = VPV^*$ . Since  $s\iota(\Gamma)s^{-1} \subseteq \iota(\Gamma)$ , every vector of  $\text{Fix}$  is fixed by each  $\pi(s\iota(\gamma)s^{-1})$ . On the other hand,  $\eta$  is fixed by all  $\pi(s\iota(\gamma)s^{-1})$  exactly when  $V^*\eta \in \text{Fix}$ . Hence  $\text{ran } P = \text{Fix} \subseteq V\text{Fix} = \text{ran } Q$ , that is,  $P \leq Q$ . The element  $r = V^*Q$  satisfies  $r^*r = Q$  and  $rr^* = V^*QV = P$ . Since  $r^*r = Q$ , we have  $r = rQ$ . Since  $\text{ran } r = \text{ran } P \subseteq \text{ran } Q$ , we also have  $r = Qr$ . Hence the cross terms in  $\sigma^* \sigma$  vanish for  $\sigma = r + (1 - Q)$ , and  $\sigma^* \sigma = Q + (1 - Q) = 1$ :  $\sigma$  is an isometry. Finiteness of  $B_\omega$  makes  $\sigma$  unitary, and  $\sigma\sigma^* = P + (1 - Q)$  forces  $Q = P$ .

*Conclusion.* Let  $\xi = [\xi_n]_\omega \in K_\omega$ , where  $\xi_n \in K_n$  is the vector of  $x_n$ ; the uniform operator-norm bound makes  $\xi$  well defined. By unitary invariance of the normalized Hilbert–Schmidt norm, the commutator hypothesis says exactly that each  $\pi(\iota(\gamma))$  fixes  $\xi$ . Thus  $\xi \in \text{Fix}$ . Since  $Q = P$ , we obtain  $V\xi \in V\text{Fix} = \text{ran } Q = \text{ran } P = \text{Fix}$ . The vector of  $U_n(s)x_n U_n(s)^*$  is  $\text{Ad } U_n(s)\xi_n$ , so, again by unitary invariance,

$$\|[U_n(s)x_n U_n(s)^*, U_n(\iota(\gamma))]\|_2 = \|(\text{Ad } U_n(\iota(\gamma)) - 1) \text{Ad } U_n(s)\xi_n\| \longrightarrow 0$$

along  $\omega$ , for every  $\gamma \in \Gamma$ . Taking  $\gamma = \gamma_0$  contradicts  $I \in \omega$ . Since the failing subsequence was arbitrary, the full sequence converges.  $\square$

*Remark 3.2* (the finite-stage proof). The ultrafilter gives no rate. Section S5 proves the theorem a second time, quantitatively at each single coordinate  $n$ . There, an explicit almost-fixed spectral subspace of the averaged adjoint action replaces the Kazhdan projection, and the equal-rank reversal estimate replaces finiteness of  $B_\omega$ . That proof makes the error bookkeeping explicit in



terms of  $\kappa$ ,  $|S|$ , the uniform bound on  $(x_n)$ , and the multiplicative defects of  $U_n$ , once the robust spectral-gap input stated there is granted at each stage.

**The finite-dimensional case** Let  $\pi: H \rightarrow \mathrm{GL}(V)$  be a finite-dimensional representation over an arbitrary field, and let

$$C = \{x \in \mathrm{End}_k(V) : \pi(\gamma)x = x\pi(\gamma) \text{ for all } \gamma \in \Gamma\}$$

be the commutant of  $\pi(\Gamma)$ . The one-sided relation  $t\Gamma t^{-1} \subseteq \Gamma$  gives  $C \subseteq uCu^{-1}$  for  $u = \pi(t)$ , and conjugation is a linear isomorphism, so the two subspaces have the same finite dimension and

$$C \subseteq uCu^{-1}, \quad \dim C = \dim uCu^{-1} \quad \implies \quad C = uCu^{-1}.$$

Since  $\pi(c) \in C$ , the element  $\pi(tct^{-1})$  again lies in  $C$  and commutes with  $\pi(\Gamma)$ , so  $\pi(u) = 1$ . This is Theorem S8.1; the argument is algebraic and uses neither property (T) nor any hypothesis on the characteristic. Applied to  $E$  it shows that every finite-dimensional linear representation over every field, and hence every homomorphism to a finite group, sends the marked word to the identity (Corollary S8.2).

Theorem 3.1 is the same principle one level of approximation up. An MF model is only asymptotically multiplicative, so its coordinate fixed spaces are not invariant subspaces; operator-norm control keeps the adjoint actions almost multiplicative on the Hilbert–Schmidt spaces, property (T) supplies an almost-fixed spectral subspace in place of  $C$ , and the equal-rank reversal estimate of Section S5 replaces equal dimension. Property (T) is what extends the conclusion from exact finite-dimensional representations to approximate operator-norm models, and it cannot be dropped: the cyclic comparison of Section S9 gives a marked group whose word is trivial in every finite-dimensional representation and nontrivial in an MF model.

#### 4 THE CENTRAL-SIGN OBSTRUCTION

Transport makes a commutator small in normalized Hilbert–Schmidt norm, and that alone obstructs nothing: padding a matrix model with an identity block leaves every operator-norm quantity unchanged and sends all normalized Hilbert–Schmidt distances to zero (Section S12). What is needed is an invariant part of the model with its own normalization. A central involution supplies one, since its negative spectral corner is again a matrix model, and on that corner the involution equals  $-1$ .

**Theorem 4.1** (central-sign criterion). *Let  $H$  and  $\Gamma$  be countable groups, let  $\Gamma$  have property (T), let  $\iota: \Gamma \rightarrow H$  be a homomorphism, and suppose*

$$t\iota(\Gamma)t^{-1} \subseteq \iota(\Gamma), \quad [c, \iota(\Gamma)] = 1.$$

*Put  $d = tct^{-1}$  and  $u = [d, \iota(a)]$  for some  $a \in \Gamma$ . If*

$$\varepsilon = u^2 \neq 1, \quad \varepsilon^2 = 1, \quad \varepsilon \in Z(H),$$

*then every homomorphism from  $H$  to a norm matrix corona maps  $\varepsilon$  to the identity, and  $H$  is not MF.*

*Proof.* Let  $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$  be a corona representation and suppose  $\Theta(\varepsilon) \neq 1$ .

*The negative corner.* Choose coordinate unitary lifts  $(V_{g,n})$  of  $\Theta$  (Lemma 2.2) and put  $p_n = \frac{1}{2}(1 + V_{\varepsilon,n})$ , the Reynolds average of the two-element subgroup  $F = \{1, \varepsilon\}$ . Because the lifts are asymptotically multiplicative and  $\varepsilon^2 = 1$ , the Hermitian part of  $p_n$  is asymptotically idempotent, so its spectrum clusters at  $\{0, 1\}$  and rounding at  $\frac{1}{2}$  by continuous functional calculus produces genuine projections at operator-norm distance  $o(1)$  from  $p_n$ ; centrality of  $\varepsilon$  makes them asymptotically central for all of  $\Theta(H)$ . Let  $q_n$  be their complements. If  $q_n$  vanished for all large  $n$  the averages would converge to 1 and  $\Theta$  would kill  $\varepsilon$ ; so  $q_n \neq 0$  along infinitely many coordinates, which we retain and relabel by  $\mathbb{N}$ . Compressing every lift to  $q_n M_{d_n}(\mathbb{C}) q_n \cong M_{r_n}(\mathbb{C})$  and polar-correcting the compressions — each is almost unitary in the corner, by asymptotic centrality — gives an operator-norm asymptotic representation  $(W_{g,n})$  of  $H$  on nonzero blocks with

$$\|W_{\varepsilon,n} + 1\| \rightarrow 0.$$

Each corner block carries its own normalized trace, so all estimates below are relative to  $r_n$  and independent of the ratio  $r_n/d_n$ .

*Transport on the corner.* Since  $c$  centralizes  $\iota(\Gamma)$ , the sequence  $(W_{c,n})$  centralizes  $(W_{\iota(\gamma),n})$  asymptotically in operator norm, hence in normalized Hilbert–Schmidt norm. The compression relation  $t\iota(\Gamma)t^{-1} \subseteq \iota(\Gamma)$  and Theorem 3.1, applied to  $x_n = W_{c,n}$ , put the conjugated sequence  $(W_{d,n})$  in the same asymptotic commutant. Therefore  $\|W_{u,n} - 1\|_2 \rightarrow 0$ , and squaring gives  $\|W_{u,n}^2 - 1\|_2 \rightarrow 0$ .

*The contradiction.* The relation  $\varepsilon = u^2$  gives  $\|W_{u,n}^2 - W_{\varepsilon,n}\| \rightarrow 0$ , while  $W_{\varepsilon,n} \rightarrow -1$  in operator norm and therefore also in normalized Hilbert–Schmidt norm. Hence  $\|W_{u,n}^2 + 1\|_2 \rightarrow 0$  and  $\|W_{u,n}^2 - 1\|_2 \rightarrow 0$ , so  $\|2 \cdot 1\|_2 \rightarrow 0$ , which is false. Thus  $\Theta(\varepsilon) = 1$ . Since  $\varepsilon \neq 1$ , no corona representation of  $H$  is injective, and  $H$  is not MF.  $\square$

Finiteness of  $F = \{1, \varepsilon\}$  entered only through the Reynolds average. Section S7 replaces  $F$  by an arbitrary finite normal subgroup of the compression defect, and then by an arbitrary normal property-(T) subgroup of it, where the negative corner becomes a moving spectral subspace. Those forms assume neither centrality nor torsion.

## 5 THE KAZHDAN–CLIFFORD CONSTRUCTION

We now build a group with the configuration of Theorem 4.1 and a nontrivial sign. Both come from the same data: a finitely presented Kazhdan group, a proper injective endomorphism of it, and an element outside the image.

With  $d = tct^{-1}$  and  $u = [d, a]$ , put

$$w = u^2.$$

Since  $d$  is an involution,  $w = [d, ada^{-1}]$ , a word in the generators, so centrality of  $w$  is a relation among them; centrality then forces  $w^2 = 1$  by (4) below.

**Theorem 5.1** (Kazhdan–Clifford construction). *Let  $\Gamma$  be a finitely presented group with property (T), let  $\alpha: \Gamma \hookrightarrow \Gamma$  be an injective endomorphism, and choose  $a \in \Gamma \setminus \alpha(\Gamma)$ . Adjoin elements  $t, c$  and impose*

$$t\gamma t^{-1} = \alpha(\gamma), \quad c^2 = 1, \quad [c, \gamma] = 1 \quad (\gamma \in \Gamma).$$

Put

$$d = tct^{-1}, \quad u = [d, a], \quad w = u^2 = [d, ada^{-1}],$$

and impose centrality of  $w$ . More explicitly, for a finite generating set  $S_\Gamma$  it is enough to impose the conjugation and commutation relations for  $s \in S_\Gamma$ , together with

$$[w, s] = 1 \quad (s \in S_\Gamma), \quad [w, t] = [w, c] = 1.$$

Thus  $w$  commutes with a generating set of the entire resulting group. The group  $E(\Gamma, \alpha, a)$  is finitely presented, the canonical map  $\Gamma \rightarrow E(\Gamma, \alpha, a)$  is injective, and  $w$  is a nontrivial central involution. Every homomorphism

$$E(\Gamma, \alpha, a) \longrightarrow \mathcal{U}(\mathcal{Q})$$

to the unitary group of every norm matrix corona maps  $w$  to 1. Consequently  $E(\Gamma, \alpha, a)$  is not MF.

*Proof.* Finite presentability follows from a relative finite presentation: start with the free product of  $\Gamma$  and the free group on  $t, c$ , and quotient by the finitely many displayed relations for a finite generating set of  $\Gamma$ .

The relation  $w^2 = 1$  is redundant. Put  $b = ada^{-1}$ . Both  $d$  and  $b$  are involutions,  $w = [d, b] = (db)^2$ , and direct cancellation gives

$$(4) \quad dwd^{-1} = w^{-1}.$$

Centrality also gives  $dwd^{-1} = w$ , whence  $w = w^{-1}$  and  $w^2 = 1$ .

For nontriviality, realize the ascending HNN extension

$$G = \langle \Gamma, t \mid t\gamma t^{-1} = \alpha(\gamma) \rangle$$

as a mapping telescope. Let  $\Gamma_\infty$  be the direct limit of the system  $\Gamma \xrightarrow{\alpha} \Gamma \xrightarrow{\alpha} \cdots$ ; injectivity of  $\alpha$  passes to the limit, so the level-zero map embeds  $\Gamma$  in  $\Gamma_\infty$ . Applying  $\alpha$  levelwise induces an injective endomorphism  $\sigma$  of  $\Gamma_\infty$ . It is surjective, hence an automorphism, because an element at level  $k$  equals  $\sigma$  of the same element at level  $k+1$ . Put  $G = \Gamma_\infty \rtimes_\sigma \mathbb{Z}$ , with stable letter  $t$  implementing  $\sigma$ ; then  $t\gamma t^{-1} = \alpha(\gamma)$  for  $\gamma \in \Gamma$ . Write  $X = G/\Gamma$  for the coset space of the level-zero copy. Let  $G$  permute the Clifford generators  $(e_x)_{x \in X}$  and map  $c$  to  $e_\Gamma$ . Then

$$d \mapsto e_{t\Gamma}, \quad ada^{-1} \mapsto e_{at\Gamma}.$$

The two cosets are distinct: equality would give  $t^{-1}at \in \Gamma$ , equivalently  $a \in \alpha(\Gamma)$ . Hence the two Clifford generators anticommute and

$$w \mapsto (e_{t\Gamma}e_{at\Gamma})^2 = -1.$$

The Clifford sign is central, so the map respects every defining relation. It follows simultaneously that  $w \neq 1$  and that the canonical map of  $\Gamma$  is injective.

Finally,  $w = u^2$  is a nontrivial central involution. By the central-sign criterion (Theorem 4.1), every corona representation maps  $w$  to the identity. Since  $w \neq 1$ , none is faithful.  $\square$

Both hypotheses on  $a$  are used, in different steps. Transport applies to  $[d, \iota(a)]$  because  $a \in \Gamma$ ; the two Clifford sites are distinct because  $a \notin \alpha(\Gamma)$ . *Remark 5.2* (a small-cancellation realization). Fournier-Facio has pointed out (personal communication, 2026) that the affine base of Sections S2–S3 can be replaced by a small-cancellation construction as in [FFF]. His finitely presented torsion-free property-(T) group  $P$  contains a copy of every finitely presented torsion-free group, in particular a direct product  $P_1 \times P_2$  with  $P_i \cong P$ . Then  $\alpha: P \xrightarrow{\cong} P_1 \leq P$ , together with any  $a \in P_2 \setminus \{1\}$ , satisfies the hypotheses of Theorem 5.1. The group  $E(P, \alpha, a)$  contains involutions whatever the base. The affine base is used for the explicit presentation of Theorem A and for Theorems C and D.

## 6 A FINITELY PRESENTED NON-MF GROUP

Specialize Theorem 5.1 to

$$(5) \quad \Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z}), \quad \alpha(v, A) = (2v, A), \quad a = (e_1, I).$$

The affine group  $\Gamma$  is finitely presented and has property (T) [BHV, Example 1.7.4(i)]; Section S2 also proves this from the presentation used below, by an exact rational sum-of-squares certificate. The endomorphism  $\alpha$  is injective because doubling is injective on  $\mathbb{Z}^3$ , it is proper because  $e_1 \notin 2\mathbb{Z}^3$ , and  $a \notin \alpha(\Gamma)$  for the same reason. Theorem 5.1 therefore applies, and relative to  $\Gamma$  the resulting group has the finite presentation

$$E = \left\langle \Gamma, t, c \left| \begin{array}{ll} t\gamma t^{-1} = \alpha(\gamma), & [c, \gamma] = 1 \quad (\gamma \in S_\Gamma), \\ c^2 = 1, & [w, g] = 1 \end{array} \right. \right\rangle,$$

where  $S_\Gamma$  is a finite generating set of  $\Gamma$ ,  $g$  runs over the generators, and  $w = [tct^{-1}, a(tct^{-1})a^{-1}]$ . Expanding a standard presentation of the affine group in the generators  $v_1, v_2, v_3$  for the translations and  $x, y, z$  for the linear part turns this into the eight-generator, forty-one-relator presentation written out in Definition S2.1; a Tietze optimization removes two generators and nine relators (Section S2).

*Proof of Theorem A.* Part (1) is Proposition S3.3.

For part (2), let  $\Theta: E \rightarrow \mathrm{U}_{\mathrm{cor}}((d_n))$  be given and pass to the corona representation  $\kappa_{(d_n)} \circ \Theta: E \rightarrow \mathcal{U}(\mathcal{Q})$  of Lemma 2.2. Apply the central-sign criterion (Theorem 4.1) with  $\Gamma = \mathcal{B}$ , the canonical homomorphism  $\iota$ , the elements  $t, c \in E$ , and  $a = v_1$ . Its hypotheses are the displayed relators: Proposition S2.2 gives property (T) of  $\mathcal{B}$ ; the six stable-letter relators give

$t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B})$ ; and the six  $c$ -commutation relators give  $[c, \iota(\mathcal{B})] = 1$ . In  $E$  we have  $d^2 = (tct^{-1})^2 = tc^2t^{-1} = 1$ , so Lemma S4.1 with  $a_1 = \iota(v_1)$  gives

$$w = [d, \iota(v_1) d \iota(v_1)^{-1}] = [d, \iota(v_1)]^2 = u^2, \quad u = [d, \iota(v_1)].$$

Centrality of  $w$  is a defining relator; with  $c^2 = 1$  it gives  $w^2 = 1$  by (4); and  $w \neq 1$  by part (1). Hence  $(\kappa_{(d_n)} \circ \Theta)(w) = 1$ , and  $\Theta(w) = 1$  because  $\kappa_{(d_n)}$  is injective.

An MF embedding of  $E$  would be an injective corona representation, and every corona representation maps the nontrivial element  $w$  to 1; so  $E$  is not MF. Suppose  $C_{\text{red}}^*(E)$  were an MF algebra, with an injective, possibly nonunital  $*$ -homomorphism  $e: C_{\text{red}}^*(E) \rightarrow \mathcal{Q}$ . Complement correction gives the injective unitary homomorphism  $u \mapsto e(u) + (1 - e(1))$ ; composing it with the canonical map  $E \rightarrow \mathcal{U}(C_{\text{red}}^*(E))$ , which is injective because  $g \mapsto \lambda_g$  is, and then with  $\kappa_{(d_n)}^{-1}$ , exhibits an injective corona representation of  $E$ . The canonical map  $E \rightarrow \mathcal{U}(C_{\text{max}}^*(E))$  is injective as well, since the left regular representation factors through  $C_{\text{max}}^*(E)$  and sends  $u_g$  to  $\lambda_g$ , so an MF embedding of  $C_{\text{max}}^*(E)$  produces an injective corona representation of  $E$  after the same conversion.  $\square$

Sections S2–S3 supply the two ingredients of this proof — property (T) of  $\mathcal{B}$  and the Clifford model witnessing  $w \neq 1$  — from the displayed relators alone, without using the identification of  $\mathcal{B}$  with the affine group.

The obstruction already appears at a finite tolerance, on a finite set of group elements, in every dimension.

**Corollary 6.1** (uniform finite operator-norm obstruction). *There are a number  $\delta > 0$  and a finite set  $F_0 \subset E$  such that, in every matrix dimension, any unitary-valued map  $\phi$  satisfying*

$$\|\phi(gh) - \phi(g)\phi(h)\| \leq \delta \quad (g, h \in F_0)$$

*also satisfies  $\|\phi(w) - 1\| < 1$ .*

*Proof.* Suppose no such pair  $(\delta, F_0)$  existed. For each  $n$ , choose a model violating the conclusion for  $\delta = 1/n$  and the  $n$ th set of an exhaustion of  $E$ . These models form an operator-norm asymptotic representation of  $E$  whose values at  $w$  stay at distance at least 1 from the identity. Its class in the unitary-sequence corona is a corona representation detecting  $w$ , contradicting Theorem A(2). The compactness argument is not quantitative; determining an effective modulus from the rational spectral certificate of Section S2 remains open.  $\square$

## 7 THE POSITION OF THE EXAMPLE

The three results of this section are proved in the supplementary part, where the machinery they use is developed.

Throughout,  $d = tct^{-1}$  and  $u = [d, \iota(v_1)]$ , so that  $w = u^2$ . Write  $\text{Rad}_{\text{MF}}(G)$  for the *MF radical* of a countable group  $G$ : the intersection of the kernels of all its corona representations (Definition S11.1). Thus  $G$  is MF exactly when

$\text{Rad}_{\text{MF}}(G) = 1$  (Proposition S11.3), and Theorem A says  $w \in \text{Rad}_{\text{MF}}(E)$  with  $w \neq 1$ .

**The quotient by the sign** Killing  $w$  does not restore the MF property.

**Theorem B** (the sign-free quotient of  $E$ ). *The quotient  $E/\langle w \rangle$  is not MF. The image of  $[\iota(v_1), d]$ , the inverse of the marked commutator  $u$ , is nontrivial there and lies in  $\text{Rad}_{\text{MF}}(E/\langle w \rangle)$ ; consequently  $\{1, w\}$  is a proper subgroup of  $\text{Rad}_{\text{MF}}(E)$ .*

The proof is in Section S11.4. It uses a second consequence of one-sided compression, proved there: if an involution is centralized by the compressed copy of a Kazhdan subgroup, and its conjugates under that subgroup commute with one another, then every corona representation identifies the entire conjugation orbit (Theorem S11.13). In  $E$  those conjugates of  $d$  commute only up to  $w$ , so in  $E/\langle w \rangle$  the element  $d$  satisfies both hypotheses, and  $u$  is killed by every corona representation there. Thus  $E/\langle w \rangle$  is a second finitely presented non-MF group, and Theorem A does not determine  $\text{Rad}_{\text{MF}}(E)$  (Section S12).

**A stably finite non-MF group algebra** Every MF algebra is stably finite [BK]; the converse fails. Here stable finiteness comes from the faithful canonical trace (Lemma S11.20), independently of the MF question. Abstract examples follow from the failure of the Connes embedding problem [FGH]; this one is explicit. For  $C_{\text{red}}^*(\text{SL}_3(\mathbb{Z}))$  and  $C_{\text{red}}^*(\text{SL}_4(\mathbb{Z}))$  the MF property is undecided; only the stronger PMF property is known to fail, for  $\text{SL}_4(\mathbb{Z})$  [MdIS].

**Theorem C** (the reduced group algebra of  $E$ ).  *$C_{\text{red}}^*(E)$  is a unital separable  $C^*$ -algebra which carries a faithful tracial state — and is therefore stably finite — but is not an MF algebra.*

*Proof.*  $C_{\text{red}}^*(E)$  is unital and separable because  $E$  is countable, and by Lemma S11.20 the canonical trace is faithful and the algebra is therefore stably finite. It is not an MF algebra by Theorem A.  $\square$

The algebra is not nuclear: the canonical map  $\mathcal{B} \rightarrow E$  is injective (Section S3), so  $E$  contains an infinite Kazhdan group and is nonamenable, whence  $C_{\text{red}}^*(E)$  is not nuclear [Lan]. The nuclear form of the Blackadar–Kirchberg problem — whether every stably finite separable nuclear  $C^*$ -algebra is quasi-diagonal — therefore remains open.

**A sofic example** Let  $X = G/\Gamma$  be the coset space of the mapping telescope  $G = \Gamma_\infty \rtimes \mathbb{Z}$  used in the proof of Theorem 5.1, at the affine data (5), and let

$$W = \text{ClLamp}(X) \rtimes G$$

be the Clifford lamp group over it. This is the target of the homomorphism used in that proof to show  $w \neq 1$ ; Section S3 realizes it over  $\mathbb{Q}$ , writing  $V$  for  $G$  in the explicit rational model.

**Theorem D** (a sofic non-MF group). *The Clifford witness  $W = \text{ClLamp}(X) \rtimes V$  of Section S3 is a finitely generated sofic group that is not MF; since sofic groups are hyperlinear,  $W$  is likewise a hyperlinear group that is not MF. In particular, neither soficity nor hyperlinearity implies the MF property.*

The proof is in Section S11. Soficity comes from the finite-index image of the doubling map: every telescope level then has finite orbits on  $X$ , so every finite subset of  $\text{ClLamp}(X) \rtimes \Gamma_\infty$  lies in a finite-by-sofic subgroup, and the remaining  $\mathbb{Z}$ -extension is sofic by Elek–Szabó [ES06]. Failure of MF follows from functoriality of the radical, since the surjection  $E \rightarrow W$  sends  $w$  to the nontrivial Clifford sign (Lemma S11.2). The same surjection shows that MF groups are not closed under quotients, and the extension  $1 \rightarrow \text{ClLamp}(X) \rightarrow W \rightarrow V \rightarrow 1$ , whose kernel is locally finite and whose quotient is residually finite, shows that they are not closed under extensions either (Section S11).

This says nothing about  $E$  itself, since  $W$  is a proper quotient of  $E$  with kernel a free group of infinite rank. We expect  $E$  to be sofic, by a block decomposition of its lamp kernel, and will give the details elsewhere. Section S12 explains why the operator-norm method decides neither soficity nor hyperlinearity for either group.



## Supplementary material

This part belongs to the same paper and uses the same bibliography. It has three jobs. Sections S1–S5 audit Theorem A: they prove the equivalence of the MF formulations, write out the literal presentation and its optimization, certify property (T) for the base by exact rational arithmetic, construct the explicit Clifford witness over  $\mathbb{Q}$ , verify the theorem directly from the forty-one relators, and reprove the transport theorem quantitatively at each single matrix dimension. Sections S6–S9 develop the mechanism in the generality it actually has. Sections S10–S13 develop the consequences and the limits of the method.

### Part I. Audit of the main theorem

#### S1 THE MF EQUIVALENCES

An *operator-norm local model* for a countable group  $G$ , a finite set  $F \subseteq G$ , a separation constant  $\delta > 0$ , and a tolerance  $\varepsilon > 0$  is a map  $V: G \rightarrow \mathcal{U}(r)$ , for some positive dimension  $r$ , such that

$$\begin{aligned} \|V(g)V(h) - V(gh)\| &\leq \varepsilon \quad (g, h \in F), \\ \|V(g) - V(h)\| &\geq \delta \quad (g \neq h \in F). \end{aligned}$$

The local formulation in Proposition 1.1 asks for such a model with  $\delta = 1$  for every finite  $F$  and every  $\varepsilon > 0$ .

*Proof of Proposition 1.1.* Lemma 2.2 identifies the unitary-sequence quotient with the unitary group of the  $C^*$ -quotient. Replacing an arbitrary finite coordinate set by a standard set of the same cardinality merely conjugates that coordinate by a permutation unitary. This proves the first equivalence once the dimension convention is normalized.

*Dimension normalization.* We first normalize dimensions, independently of everything below. Given positive dimensions  $d_n$ , replace the  $n$ th coordinate model by

$$\tilde{V}_n(g) = I_{X_0} \oplus \cdots \oplus I_{X_{n-1}} \oplus V_n(g)$$

on  $X_0 \oplus \cdots \oplus X_n$ , of strictly increasing dimension  $D_n = d_0 + \cdots + d_n$ . Products, multiplicative defects, and pairwise operator-norm distances are unchanged, because block-diagonal operator norms are maxima and every earlier block is the same identity for all group elements; no approximation error from earlier coordinates is carried forward. The construction therefore preserves and reflects the quotient relation and injectivity, and forgetting strict increase proves the converse. The universal-quotient construction of Proposition S11.3 uses exactly this step to meet the strictly-increasing dimension convention; nothing below depends on that proposition.

Suppose first that  $\rho: G \rightarrow \mathcal{U}_{\text{cor}}((d_n))$  is injective, and choose unitary lifts  $u_n(g)$  of  $\rho(g)$ . Fix a finite set  $F \subseteq G$  and  $\varepsilon > 0$ . For each ordered pair  $g \neq h$  in  $F$ , injectivity says that  $(u_n(g)^{-1}u_n(h))_n$  does not converge to the identity

in operator norm. Hence there are  $\delta_{g,h} > 0$  and arbitrarily large coordinates at which

$$\|u_n(g) - u_n(h)\| \geq \delta_{g,h}.$$

Choose an integer  $N_{g,h}$  with  $8 < N_{g,h} \delta_{g,h}^2$ . An explicit tensor-power estimate then applies: whenever two unitaries are  $\delta_{g,h}$ -separated, some tensor power of exponent  $p \leq N_{g,h}$  separates them by more than 1 — amplifying a spectral gap of the relative unitary until it crosses the unit circle takes at most  $N_{g,h}$  steps once  $8 < N_{g,h} \delta_{g,h}^2$ . The quotient homomorphism law makes all multiplicative defects for pairs in  $F$  arbitrarily small at every sufficiently large coordinate. Choose, for each  $g \neq h$ , a coordinate with the displayed separation and with every required defect at most  $\varepsilon/N_{g,h}$ . Then the  $p$ th tensor power separates the designated pair by more than 1, while the tensor-product telescoping identity bounds each new defect by  $p\varepsilon/N_{g,h} \leq \varepsilon$ . Finally take the block sum of these finitely many pair-specific models. Operator norms of block diagonal matrices are maxima, so every pair in  $F$  is separated by at least 1 and every required product has defect at most  $\varepsilon$ . This gives the normalized local model.

Conversely, local models with separation constant 1 for all finite sets and tolerances make the group *weakly* MF. Choose models along an exhaustion with errors tending to zero. Then each fixed pair  $g, h$  is eventually represented multiplicatively, and each fixed pair  $g \neq h$  is eventually separated by 1. Hence the classes define an injective homomorphism into the unitary-sequence quotient, with no condition on the dimension sequence. Weak MF in turn makes the norm-MF radical of Definition S11.1 trivial, and the block-sum diagonalization in the proof of Proposition S11.3(2) then assembles the detecting sequences into a single faithful corona representation. This weak-to-corona implication is thus obtained from the universal-quotient construction — whose dimension bookkeeping is the normalization step above — not from a separate exhaustion.  $\square$

## S2 THE EXPLICIT PRESENTATION AND ITS PROPERTY-(T) BASE

We work throughout with the groups defined by the relators below. Classically the base is the affine group  $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ , a familiar Kazhdan group (Remark S2.4). The proof of property (T) given here nevertheless uses only the displayed presentations and is independent of the identification: an exact rational spectral certificate for an auxiliary Steinberg-type presentation, an explicit surjection onto the linear presentation, and a fixed-point bridge from the linear subgroup to the full base. First define the *linear presentation*

$$(6) \quad \mathcal{R} = \left\langle x, y, z \left| \begin{array}{l} x^3 = y^3 = z^2 = (xz)^3 = (yz)^3 = 1, \\ (x^{-1}zxy)^2 = (y^{-1}zyx)^2 = (xy)^6 = 1 \end{array} \right. \right\rangle,$$

and the *base*

(7)

$$\mathcal{B} = \left\langle v_1, v_2, v_3, x, y, z \left| \begin{array}{l} \text{the eight relations in (6),} \\ [v_1, v_2] = [v_1, v_3] = [v_2, v_3] = 1, \\ xv_1x^{-1} = v_3, \quad xv_2x^{-1} = v_1, \quad xv_3x^{-1} = v_2, \\ yv_1y^{-1} = v_1, \quad yv_2y^{-1} = v_2^{-1}v_3, \quad yv_3y^{-1} = v_1v_2^{-1}, \\ zv_1z^{-1} = v_2v_3^{-1}, \quad zv_2z^{-1} = v_1v_3^{-1}, \quad zv_3z^{-1} = v_3^{-1} \end{array} \right. \right\rangle.$$

Write  $\iota: \mathcal{B} \rightarrow E$  for the homomorphism induced by the first six generators of the presentation below. This map is not assumed to be injective.

**Definition S2.1.** Use the six base generators and the relations displayed in (7). Define

$$(8) \quad E = \left\langle v_1, v_2, v_3, x, y, z, t, c \left| \begin{array}{l} \text{all relations in (7),} \\ tv_it^{-1} = v_i^2 \quad (1 \leq i \leq 3), \\ txt^{-1} = x, \quad tyt^{-1} = y, \quad tzt^{-1} = z, \\ c^2 = 1, \\ [c, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z\}), \\ [w, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z, t, c\}) \end{array} \right. \right\rangle,$$

where  $w$  abbreviates the word  $[tct^{-1}, v_1(tct^{-1})v_1^{-1}]$ . This is an explicit finite presentation with eight generators and forty-one relators: the twenty base relators, six stable-letter relators,  $c^2$ , the six relations making  $c$  commute with the base generators, and eight centrality relators. Its defining relations make  $w$  central, and centrality together with  $c^2 = 1$  implies  $w^2 = 1$  by (4). The relations on the six base generators also give

$$t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B}), \quad [c, \iota(\mathcal{B})] = 1.$$

Six generators suffice. The eight-generator presentation is used because it displays the structure used in the proof: the three commuting translations, the linear generators acting on them, the stable letter  $t$ , and  $c$ . The group  $E$  is nevertheless generated by the six elements  $v_1, x, y, z, t, c$ . There is also a Tietze-equivalent presentation on these six generators with thirty-two relators. In (8), eliminate  $v_3 = xv_1x^{-1}$  and  $v_2 = x^2v_1x^{-2}$  by Tietze substitutions and delete the two defining relations; the remaining  $x$ -action relation follows from  $x^3 = 1$ , and the stable-letter,  $c$ -commutation, and centrality relations for  $v_2, v_3$  follow from those for  $v_1$  together with  $[t, x] = 1$  and  $x^3 = 1$ . Deleting these nine leaves thirty-two relators on  $v_1, x, y, z, t, c$ .

Figure 1 isolates the geometry used later in the analytic argument: the relations force  $d = tct^{-1}$  to commute with the compressed copy  $t\iota(\mathcal{B})t^{-1}$ , and they do not constrain  $[d, \iota(v_1)d\iota(v_1)^{-1}]$ .

## S2.1 Property (T) of the base

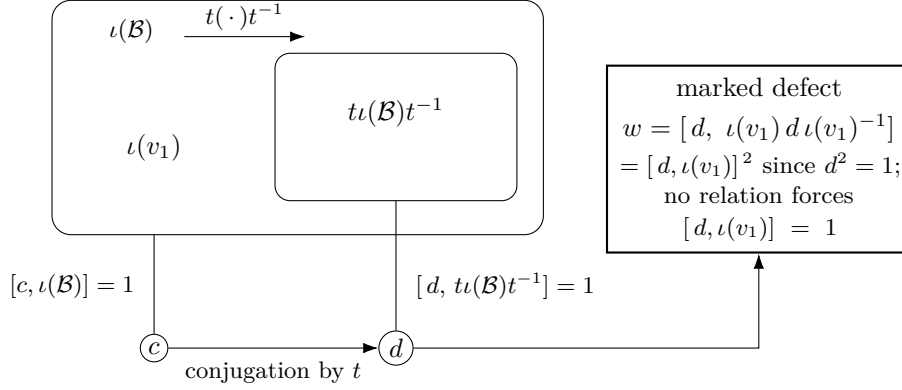


Figure 1: The one-sided conjugation configuration. Conjugation by  $t$  carries the image of the base into itself. It sends  $c$  to  $d = tct^{-1}$ : the relations force  $c$  to commute with all of  $\iota(\mathcal{B})$  and force  $d$  to commute with the compressed copy, but do not require  $d$  to commute with  $\iota(v_1)$ . The marked word  $w$  is that commutation defect. Only the one-sided inclusion is used by the analytic criterion.

**Proposition S2.2.** *The twenty-relator group  $\mathcal{B}$  has property (T), in both the real-orthogonal and complex-unitary formulations.*

*Proof. Step 1: a certified Steinberg-type presentation.* Let  $P_{13}$  be the group on the six generators  $e_{12}, e_{13}, e_{21}, e_{23}, e_{31}, e_{32}$  with the thirteen relators

$$\begin{aligned} [e_{12}, e_{23}] &= e_{13}, & [e_{13}, e_{32}] &= e_{12}, & [e_{21}, e_{13}] &= e_{23}, \\ [e_{23}, e_{31}] &= e_{21}, & [e_{31}, e_{12}] &= e_{32}, & [e_{32}, e_{21}] &= e_{31}, \\ [e_{12}, e_{13}] &= [e_{12}, e_{32}] = [e_{13}, e_{23}] = [e_{21}, e_{23}] = [e_{21}, e_{31}] = [e_{31}, e_{32}] = 1, \\ (e_{12} e_{21}^{-1} e_{12})^4 &= 1, \end{aligned}$$

the elementary-matrix relations in rank three. An exact sum-of-squares certificate with integer matrix data, verified by denominator-free rational arithmetic, proves the degree-one Hodge estimate for  $P_{13}$  with gap  $\frac{1}{500}$ . Testing it on coboundaries gives, in every orthogonal representation, the quadratic gap

$$\|\Delta\xi\|^2 \geq \frac{1}{500} \langle \Delta\xi, \xi \rangle$$

for the six-generator Laplacian  $\Delta$ . The gap places the spectrum of  $\Delta$  in  $\{0\} \cup [\frac{1}{500}, \infty)$ . The kernel of  $\Delta$  is exactly the space of invariant vectors. The localization makes 0 an isolated point of the spectrum but does not by itself exclude it, so one more step is needed: in a representation with no nonzero invariant vector the kernel is trivial, and  $\Delta$  is then bounded below. Hence a unit vector almost fixed by the six generators has small Laplacian energy and is close to an invariant vector. This yields a Kazhdan pair:  $P_{13}$  has property (T). Certificates of this kind go back to Ozawa's

algebraic characterization of property (T) [Oz] and its computer-assisted implementations for  $\mathrm{SL}_3(\mathbb{Z})$  [NT].

*Step 2: the explicit surjection onto the linear presentation.* Map the six generators to the words

$$\begin{aligned} e_{12} &\mapsto y^{-1}xyz^{-1}x^{-1}, & e_{13} &\mapsto xzy^{-1}x^{-1}y, & e_{21} &\mapsto y^{-1}xyxz, \\ e_{23} &\mapsto x^{-1}y^{-1}xyz^{-1}, & e_{31} &\mapsto xy^{-1}xyz^{-1}x, & e_{32} &\mapsto x^{-1}zy^{-1}x^{-1}y^{-1}x^{-1} \end{aligned}$$

in the linear presentation  $\mathcal{R}$  of (6). A word-by-word verification inside  $\mathcal{R}$ , using only its eight displayed relators, checks that all thirteen relators of  $P_{13}$  map to the identity, so the assignment defines a homomorphism  $P_{13} \rightarrow \mathcal{R}$ . The same computation exhibits explicit words in the six images whose values are  $z$  and  $xy$ . A further computation shows that  $z$  and  $xy$  generate  $\mathcal{R}$ . Hence the homomorphism is surjective. Property (T) passes to quotients, so  $\mathcal{R}$  has property (T).

*Step 3: the fixed-point bridge to the base.* The generators  $x, y, z$  of  $\mathcal{B}$  satisfy the relators of  $\mathcal{R}$ , so  $\mathcal{R}$  maps to the linear subgroup of  $\mathcal{B}$  they generate, and Step 2 gives a finite set  $S$  of words in  $x, y, z$  and a  $\kappa > 0$  forming a Kazhdan pair for it. Let  $\rho$  be an orthogonal representation of  $\mathcal{B}$  with a unit vector  $\xi$  displaced by less than  $\kappa/64$  by every element of the finite control set consisting of  $S$  together with the translations  $v_2, v_3$ . The third basis translation needs no separate control:  $v_1 = xv_2x^{-1}$  by the displayed  $x$ -action, and conjugation by the fixed unitary of  $x$  preserves displacements of vectors fixed by the linear subgroup. Three estimates produce an invariant vector near  $\xi$ .

- (1) Projecting  $\xi$  to the subspace of vectors fixed by the linear subgroup gives such a fixed vector  $p$  with  $\|p - \xi\| < 1/64$ , by the moving-projection estimate for the Kazhdan pair.
- (2) Translations move  $p$  by a bound independent of the lattice vector. Three explicit words in  $x, y, z$  act on the translation lattice as elementary transvections. Consequently every lattice element is a product of two linear-subgroup conjugates of basis translations — a two-conjugate normal form proved inside the presentation. A conjugate  $rvr^{-1}$  with  $r$  in the linear subgroup displaces the fixed vector  $p$  exactly as much as  $v$  does. Each control translation displaces  $p$  by less than  $3/64$ , because its displacement bound at  $\xi$  transfers across  $\|p - \xi\|$ . Hence every translation displaces  $p$  by at most  $1/8$ .
- (3) The orbit of  $p$  under the translation subgroup is bounded, so the Hilbert-space circumcenter argument gives a translation-fixed vector  $q$  with  $\|q - p\| \leq 1/8$ ; the orthogonal projection  $\eta$  of  $p$  to the translation-fixed subspace is the nearest fixed vector, so  $\|\eta - p\| \leq \|q - p\| \leq 1/8$ . The linear subgroup normalizes the translation subgroup and fixes  $p$ , so  $\eta$  is fixed by it as well. Translations and the linear generators generate  $\mathcal{B}$ , so  $\eta$  is  $\mathcal{B}$ -invariant. Finally,  $\|\eta - \xi\| \leq 1/8 + 1/64 < 1$  keeps  $\eta$  nonzero.

Thus almost-invariant vectors produce invariant vectors and  $\mathcal{B}$  has property (T) in the real-orthogonal formulation; realification and complexification identify it with the complex-unitary formulation.  $\square$

The auxiliary presentation of Step 1 presents the group it is modelled on.

**Theorem S2.3** (completeness of the thirteen-relator presentation). *The canonical homomorphism  $P_{13} \rightarrow \mathrm{SL}_3(\mathbb{Z})$ , sending each generator  $e_{ij}$  to the corresponding elementary matrix, is an isomorphism.*

*Proof route.* Surjectivity is Euclidean elimination. Injectivity is the unstable  $K_2(3, \mathbb{Z}) = \mathbb{Z}/2$  computation in presentation form: every word in unit root letters normalizes, by a descent on the lexicographic measure of its trajectory over the third basis column, into a norm-monotone word times an element of the signed-swap subgroup  $\langle w_{13}, w_{23} \rangle$ . For a kernel word the monotone trajectory is squeezed at norm one, so every letter fixes the base column; conjugating through the signed-swap tail lands the word in the parabolic subgroup  $\langle e_{12}, e_{21}, e_{31}, e_{32} \rangle$ , whose elements the matrix model separates by their semidirect normal form. The signed-swap subgroup itself is governed, over  $\mathbb{Z}$ , by a six-coset Schreier system whose every Schreier element is a power of the block Weyl element; the classical obstruction  $\{-1, -1\} = w^4$  is killed by the displayed relator, and with it the kernel.  $\square$

*Remark S2.4* (the classical identification). The eight relations (6) present  $\mathrm{SL}_3(\mathbb{Z})$ : Theorem S2.3 composed with the surjection of Step 2 identifies  $\mathcal{R}$  with  $\mathrm{SL}_3(\mathbb{Z})$ , and the base  $\mathcal{B}$  is then the affine group  $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$  — the surjection  $\mathcal{B} \rightarrow \bar{\Gamma}$  of Lemma S3.2 onto the matrix realization is an isomorphism. The presentations go back to [CRW, Theorem 2], with the matrix-labelled generators also recorded in [CLV], and property (T) of the affine group is classical [BHV, Example 1.7.4(i)]; the proofs here are independent of these sources, and so is the identification.

### S3 THE WITNESS: $w \neq 1$

Clifford signs, shifts, and HNN doubling have appeared together before in approximate-representation theory, notably in Slofstra’s hyperlinear-profile construction [Slo18, Section 2]. The construction below uses the Clifford lamp as an explicit algebraic witness.

**Construction S3.1** (Clifford lamp groups). For a set  $X$ , use the real Clifford-algebra convention generated by  $(c_x)_{x \in X}$  with  $c_x^2 = 1$  and  $c_x c_y = -c_y c_x$  for  $x \neq y$ , and let  $\mathrm{ClLamp}(X)$  be the subgroup of units generated by  $-1$  and the  $c_x$ . Write  $\zeta = -1$ . In this concrete group,

$$\zeta^2 = c_x^2 = 1, \quad c_x c_y = \zeta c_y c_x \quad (x \neq y).$$

The sign  $\zeta$  is nontrivial. Thus  $\mathrm{ClLamp}(X)$  is a central extension, by the sign  $\zeta$ , of the elementary abelian 2-group on the sites  $X$ : distinct site involutions commute only up to  $\zeta$ , unlike the commuting lamps of a classical lamplighter. Every permutation of  $X$  acts on this group by  $c_x \mapsto c_{\sigma(x)}$  and fixes  $\zeta$ .

**Lemma S3.2** (linear model). *For  $M \in \mathrm{GL}_3(\mathbb{Q})$  and  $u \in \mathbb{Q}^3$ , write*

$$A(M, u) = \begin{pmatrix} M & u \\ 0 & 1 \end{pmatrix}.$$

*Assign  $v_i \mapsto A(I_3, e_i)$  and*

$$\begin{aligned} x &\mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, 0\right), & y &\mapsto A\left(\begin{pmatrix} 1 & 0 & 1 \\ 0 & -1 & -1 \\ 0 & 1 & 0 \end{pmatrix}, 0\right), \\ z &\mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \end{pmatrix}, 0\right). \end{aligned}$$

*Let  $\bar{\Gamma} \leq \mathrm{GL}_4(\mathbb{Q})$  be the subgroup generated by these six matrices, and put  $D = \mathrm{diag}(2, 2, 2, 1)$ . Direct matrix computation proves that the six matrices satisfy all twenty relators of  $\mathcal{B}$  and that*

$$\bar{\alpha}(u) = DuD^{-1}$$

*is an injective endomorphism of  $\bar{\Gamma}$  which squares the three translation generators and fixes  $x, y, z$ . Moreover the matrix assigned to  $v_1$  does not lie in  $\mathrm{range}(\bar{\alpha})$ .*

*Proof.* The twenty relator checks reduce to sixteen rational identities for each matrix equation. Conjugation by  $D$  gives the six stable-letter equations and is injective because it is conjugation by an invertible matrix. Every displayed generator and its explicitly computed inverse has integral entries; equivalently, all six generators lie in  $\mathrm{GL}_4(\mathbb{Z})$ . Hence  $\bar{\Gamma} \leq \mathrm{GL}_4(\mathbb{Z})$ . However  $D^{-1}\bar{v}_1D$  has entry  $1/2$  in position  $(1, 4)$ , so it is not in  $\bar{\Gamma}$ . This proves the last assertion.  $\square$

Form the direct limit  $T = \varinjlim (\bar{\Gamma}, \bar{\alpha})$ , the algebraic mapping telescope of  $\bar{\alpha}$ , extend it by its shift  $\tau$ , and call the resulting group  $V = T \rtimes \mathbb{Z}$ . Let  $j: \bar{\Gamma} \hookrightarrow V$  be the level-zero map. Put  $X = V/j(\bar{\Gamma})$ , with root  $o = j(\bar{\Gamma})$ . The final assertion of Lemma S3.2 states that  $\tau o \neq j(\bar{v}_1)\tau o$ ; the Clifford generators at these two sites therefore anticommute.

**Proposition S3.3.**  *$w \neq 1$  in  $E$ .*

*Proof.* Form  $W := \mathrm{ClLamp}(X) \rtimes V$  as in Construction S3.1. Define a map on the generators of (8) by

$$\begin{aligned} g &\mapsto j(\bar{g}) \in V \quad (g \in \{v_1, v_2, v_3, x, y, z\}), \\ t &\mapsto \tau, \quad c \mapsto c_o. \end{aligned}$$

The exact checks in Lemma S3.2 give the twenty base and six stable-letter relations. The level-zero subgroup fixes  $o$ , so  $c_o$  commutes with every base generator, and  $c_o^2 = 1$ . Under this assignment,

$$tct^{-1} \mapsto c_{\tau o}, \quad v_1(tct^{-1})v_1^{-1} \mapsto c_{j(\bar{v}_1)\tau o},$$



and the two sites are distinct by Lemma S3.2, so

$$w \mapsto [c_{\tau o}, c_{j(\bar{v}_1)\tau o}] = -1 = \zeta.$$

Since  $\zeta$  is central in  $W$ , the remaining relations  $[w, g] = 1$  for  $g \in \{v_1, v_2, v_3, x, y, z, t, c\}$  are also satisfied. Hence the assignment extends to a group homomorphism  $E \rightarrow W$  sending  $w$  to  $\zeta \neq 1$ , and therefore  $w \neq 1$  in  $E$ .  $\square$

When  $X$  is infinite so is  $\text{ClLamp}(X)$ , but each of its elements involves only finitely many Clifford sites, and the remaining data of the target are rational matrices. The two sites are separated by the range characterization of Lemma S3.2:  $D^{-1}\bar{v}_1 D$  has the entry  $1/2$  in position  $(1, 4)$  and therefore does not lie in  $\bar{\Gamma} \leq \text{GL}_4(\mathbb{Z})$ .

#### S4 THE DEFECT SQUARE

**Lemma S4.1** (the marked word is a defect square). *Let  $H_1$  be a group and  $d, a_1 \in H_1$  with  $d^2 = 1$ . Then*

$$[d, a_1 d a_1^{-1}] = [d, a_1]^2.$$

*Proof.* Put  $b = a_1 d a_1^{-1}$ , so  $b^2 = a_1 d^2 a_1^{-1} = 1$ ; thus  $b^{-1} = b$  and  $d^{-1} = d$ . Then  $[d, a_1] = d a_1 d^{-1} a_1^{-1} = d(a_1 d a_1^{-1}) = db$ , and

$$[d, a_1]^2 = (db)^2 = db db = d b d^{-1} b^{-1} = [d, b]. \quad \square$$

The marked radical element. The element  $w$  lies in the kernel of *every* corona representation of  $E$ , not merely in the kernel of each one separately, so every corona representation factors through  $E/\langle w \rangle$ ; here  $\langle w \rangle = \{1, w\}$  is central, hence normal. In the language of approximation radicals,  $w$  is a nontrivial element of the MF radical of  $E$ . The terminology follows Slofstra's quotients  $G^{\text{fin}}$  and  $G^{\text{fa}}$ , obtained by killing the elements invisible to exact and to approximate finite-dimensional representations [Slo17, Definitions 2.5–2.7]. Theorem S7.3 extends the conclusion to every countable group with a Kazhdan conjugation datum: every finite normal subgroup of the defect subgroup lies in the MF radical. Its proof gives  $\|U_{u,n} - 1\|_2 \rightarrow 0$  for the unsquared commutator  $u = [d, \iota(v_1)]$  in every operator-norm almost representation of  $E$ , and Theorem B places  $u$  itself in the radical:  $\text{Rad}_{\text{MF}}(E)$  properly contains  $\{1, w\}$ .

Corner ranks. The coordinate Reynolds corner in the proof of Theorem S7.3 discards the zero-rank coordinates and equips each retained corner block  $q_n M_{d_n}(\mathbb{C}) q_n$  with its own normalized trace, so every Hilbert–Schmidt estimate there is relative to  $r_n = \text{rank } q_n$  and independent of the ratio  $r_n/d_n$ . Small relative ranks therefore cost nothing in the operator-norm setting, whereas they block the same argument in Hilbert–Schmidt metrics (Section S12).

## S5 THE FINITE-STAGE TRANSPORT PROOF

This appendix proves Theorem 3.1 a second time, quantitatively at each single coordinate  $n$ : every estimate below is an explicit inequality in  $\kappa$ ,  $|S|$ , the uniform bound on  $(x_n)$ , and the multiplicative defects of  $U_n$ , once the robust spectral-gap input of (i) is granted at each stage. That input is the one place where a limit still enters, as Remark 3.2 says: property (T) constrains exact unitary representations, while at a finite stage the microstates only generate a representation of the free group on  $S$ , so no threshold in  $\kappa$ ,  $|S|$  and the defects on a fixed finite set can supply it. Making that step effective as well would need a sum-of-squares certificate for property (T) in the sense of [Oz]. Corollary S7.4 and Section S12 refer to this proof.

*Finite-stage proof of Theorem 3.1.* Property (T) provides a finite symmetric generating set  $S \subseteq \Gamma$  with  $1 \in S$  and a Kazhdan constant  $\kappa \in (0, 1]$  for the pair  $(S, \kappa)$ . The adjoint model is that of the printed proof: each group element acts by conjugation,  $\text{Ad } U_n(g) \xi = U_n(g) \xi U_n(g)^*$ , on  $L^2(M_{d_n}(\mathbb{C}), \text{tr}_{d_n})$ , and operator-norm almost multiplicativity of  $U_n$  makes  $g \mapsto \text{Ad } U_n(g)$  an operator-norm asymptotic representation on these Hilbert spaces.

*The almost-fixed spectral subspace.* Let  $H_n$  be the Hermitian part of the average  $\frac{1}{|S|} \sum_{s' \in S} \text{Ad } U_n(\iota(s'))$ , fix the threshold

$$\theta = \frac{1}{2} \left( \left( 1 - \frac{\kappa^2}{4|S|} \right) + 1 \right) < 1,$$

and let  $P_n$  be the spectral projection of  $H_n$  for the part of the spectrum above  $\theta$ . Two spectral estimates tie  $P_n$  to the group.

- (i) *Displacement.* For every  $\gamma \in \Gamma$ ,  $\|(\text{Ad } U_n(\iota(\gamma)) - 1) P_n\| \rightarrow 0$ . The mechanism is a robust spectral gap: for every pair  $a, b$  with  $1 - \kappa^2/(4|S|) < a$  and  $b < 1$ , once the multiplicative defects of  $U_n$  on the finitely many group elements involved are small enough, no eigenvalue of  $H_n$  lies in  $[a, b]$  — an eigenvector with intermediate eigenvalue would be a unit vector that the Kazhdan pair forbids: too fixed to be moved by  $\kappa$ , too moved to be fixed. Hence the spectrum of  $H_n$  above  $\theta$  is eventually within any prescribed  $\delta$  of 1, so a unit vector  $\eta \in \text{ran } P_n$  has  $\langle (1 - H_n)\eta, \eta \rangle \leq \delta$ ; since  $1 - H_n$  is the average of the positive operators  $1 - \text{Re Ad } U_n(\iota(s'))$ , each generator satisfies  $\frac{1}{2} \|(\text{Ad } U_n(\iota(s')) - 1)\eta\|_2^2 \leq |S| \delta$ , and a word in  $S$  of length  $\ell$  moves  $\eta$  by at most  $\ell\sqrt{2|S|\delta}$  plus the accumulated multiplicative defects. The projection  $P_n$  is the finite-stage substitute for the Kazhdan projection.
- (ii) *Spectral concentration.* Write  $\xi_n \in L^2(M_{d_n}(\mathbb{C}), \text{tr}_{d_n})$  for the vector  $x_n$ . The commutator hypothesis says each  $\text{Ad } U_n(\iota(s'))$  almost fixes  $\xi_n$ , so the residual  $(1 - H_n)\xi_n$  is Hilbert–Schmidt null. On  $\text{ran}(1 - P_n)$  the operator  $1 - H_n$  is bounded below by  $1 - \theta$ , and it commutes with  $P_n$ , so every vector  $\eta$  satisfies

$$\|(1 - P_n)\eta\|_2 \leq (1 - \theta)^{-1} \|(1 - H_n)\eta\|_2.$$

Applied to  $\eta = \xi_n$ , this shows  $\xi_n$  is asymptotically concentrated in  $\text{ran } P_n$ .

*Reversing the one-sided projection estimate.* Put  $T_n = \text{Ad } U_n(s)$  and  $Q_n = T_n P_n T_n^*$ , a projection of the same rank as  $P_n$ . The conjugation relation enters exactly once: for  $\gamma \in \Gamma$  the element  $s^{-1}$  pulls the almost-fixedness of a vector of  $\text{ran } P_n$  under  $\iota(\Gamma) \supseteq s\iota(\Gamma)s^{-1}$  back to almost-fixedness under each  $\iota(s')$ ,  $s' \in S$ . By the concentration estimate, the pulled-back vector is again almost in the subspace. Thus  $\text{ran } P_n$  is almost contained in its conjugate,  $\|(1 - Q_n)P_n\| \rightarrow 0$  — the finite-stage form of the one-sided inclusion  $P \leq VPV^*$ . Since  $\text{rank } P_n = \text{rank } Q_n$ , the equal-rank reversal estimate — for projections  $p, q$  of equal finite rank,  $\|(1 - q)p\| \leq \varepsilon < 1$  implies  $\|(1 - p)q\| \leq \varepsilon/\sqrt{1 - \varepsilon^2}$  — reverses the containment:

$$\|(1 - P_n)Q_n\| \rightarrow 0.$$

This is the middle implication of the display at the head of Section 3, and it is the finite-dimensional substitute for finiteness of a norm ultraproduct: a one-sided compression cannot enlarge the subspace.

*Combining the estimates.* Fix  $\gamma$  and  $\varepsilon > 0$ , let  $M \geq 1$  bound  $\|x_n\|$ , and put  $q = \min(1, \varepsilon/(40(M^2 + 1)))$ . For all large  $n$  the three estimates hold at tolerance  $q$ :

$$\|(1 - P_n)\xi_n\|_2^2 \leq q, \quad \|(1 - P_n)Q_n\| \leq q, \quad \|(A_n - 1)P_n\| \leq q,$$

where  $A_n = \text{Ad } U_n(\iota(\gamma))$ . Decompose the transported vector along  $P_n$  and  $Q_n$ :

$$T_n \xi_n = P_n Q_n T_n \xi_n + (1 - P_n)Q_n T_n \xi_n + (1 - Q_n)T_n \xi_n.$$

Since  $Q_n = T_n P_n T_n^*$  and  $T_n$  is conjugation by a unitary,

$$\|(1 - Q_n)T_n \xi_n\|_2 = \|T_n(1 - P_n)\xi_n\|_2 = \|(1 - P_n)\xi_n\|_2 \leq q^{1/2};$$

the middle term obeys  $\|(1 - P_n)Q_n T_n \xi_n\|_2 \leq \|(1 - P_n)Q_n\| \|\xi_n\|_2 \leq qM$ , and the displacement estimate moves the first term by at most  $q \|\xi_n\|_2 \leq qM$ . Applying  $A_n - 1$ , whose operator norm is at most 2, to the last two terms and using the displacement bound on the first gives

$$\|(A_n - 1)T_n \xi_n\|_2^2 \leq (qM + 2qM + 2q^{1/2})^2 \leq 18q^2 M^2 + 16q \leq \varepsilon.$$

Since  $T_n \xi_n$  is the vector of  $U_n(s)x_n U_n(s)^*$ , this proves the asserted convergence — with an explicit rate in terms of  $\kappa$ ,  $|S|$ ,  $M$ , and the multiplicative defects of  $U_n$  on the finitely many group elements involved.  $\square$

## Part II. The mechanism in general

The main text used one transport theorem, one corner, and one finite central subgroup. Each is a special case. Transport holds at every Hilbertian scale and for intertwiners between different models; the corner argument holds for every finite normal subgroup of the compression defect, and then for every normal property-(T) subgroup of it, with no centrality and no torsion; and the whole configuration has an exact finite-dimensional shadow that needs no analysis at all. A cyclic comparison closes the section by showing that property (T) cannot be dropped.

### S6 TRANSPORT AT EVERY SCALE

The transport theorem holds at every Hilbertian scale, not only at the dimension normalization.

**Theorem S6.1** (transport at every scale, and of intertwiners). *Let  $\Gamma$ ,  $H$ ,  $\iota$ , and  $s$  be as in Theorem 3.1.*

- (1) *Let  $(U_n)$  be as there, let  $(w_n)$  be nonnegative weights, and suppose there is  $C$  with  $\text{Tr}(x_n^* x_n) \leq C w_n$  for all  $n$ , and that for every  $\gamma \in \Gamma$  and every  $\varepsilon > 0$ , eventually*

$$\text{Tr} |x_n - U_n(\iota(\gamma)) x_n U_n(\iota(\gamma))^*|^2 \leq \varepsilon w_n.$$

*Then  $U_n(s) x_n U_n(s)^*$  and  $U_n(s)^* x_n U_n(s)$  satisfy both conditions as well.*

- (2) *Let  $U_n^1: H \rightarrow \mathcal{U}(d_n^1)$  and  $U_n^2: H \rightarrow \mathcal{U}(d_n^2)$  be operator-norm asymptotic representations and let  $(x_n)$  be uniformly operator-norm bounded  $d_n^1 \times d_n^2$  matrices with*

$$\frac{\text{Tr} |x_n - U_n^1(\iota(\gamma)) x_n U_n^2(\iota(\gamma))^*|^2}{d_n^1 + d_n^2} \longrightarrow 0 \quad (\gamma \in \Gamma).$$

*Then  $U_n^1(s) x_n U_n^2(s)^*$  satisfies the same convergence for every  $\gamma \in \Gamma$ .*

*Proof.* (1) Give each coordinate space with  $w_n > 0$  the inner product  $\langle x, y \rangle = \text{Tr}(y^* x)/w_n$ ; an index with  $w_n = 0$  forces  $x_n = 0$ . Conjugation by a unitary is unitary for every weight, and both hypotheses and both conclusions concern only the renormalized norms, so the ultraproduct proof of Theorem 3.1 applies verbatim: the mass bound defines the ultraproduct vector, the adjoint actions remain operator-norm almost multiplicative, and finiteness reverses the one-sided compression in both directions.

(2) The block sum  $W_n = U_n^1 \oplus U_n^2$  is an operator-norm asymptotic representation, since block-diagonal operator norms are maxima. The off-diagonal embedding of  $x_n$  has operator norm  $\|x_n\|$ , its commutator with  $W_n(\iota(\gamma))$  is the off-diagonal embedding of the intertwining defect — whose normalized Hilbert–Schmidt norm on the block space carries the denominator  $d_n^1 + d_n^2$  — and conjugation by  $W_n(s)$  acts on the embedded corner by  $x_n \mapsto U_n^1(s) x_n U_n^2(s)^*$ . Theorem 3.1 applied to  $W_n$  transports the embedded

sequence, and the same reduction through (1) gives the intertwiner transport at every weight.  $\square$

The dimension weight  $w_n = d_n$  recovers Theorem 3.1 with the uniform operator-norm bound relaxed to mass boundedness; the collapse proof of Theorem S11.13 uses a rank weight, which normalizes a defect at its own scale; and the adjoint sector of Theorem 3.1 is the diagonal case of (2).

Fixed tensors of type  $(p, q)$  are Hilbert–Schmidt intertwiners from the  $q$ th to the  $p$ th tensor power. Tensor powers of unitaries are exactly multiplicative and satisfy  $\|A^{\otimes p} - B^{\otimes p}\| \leq p \|A - B\|$ , so tensor powers of an operator-norm asymptotic representation are again operator-norm asymptotic representations, and Theorem S6.1(2) applies to every pair of them: one-sided compression of a Kazhdan image creates no asymptotic invariant tensors of any fixed type, at any weight.

## S7 THE COMPRESSION DEFECT AND ITS OBSTRUCTION THEOREMS

The transport theorem has a group-theoretic output. We fix the configuration it needs once and for all.

**Definition S7.1** (Kazhdan conjugation datum). Let  $H$  be a countable group. A *Kazhdan conjugation datum* in  $H$  is a tuple  $(\Lambda, \iota, t, c)$ , where  $\Lambda$  is a countable group with property (T),  $\iota: \Lambda \rightarrow H$  is a group homomorphism, not assumed injective, and  $t, c \in H$ , subject to

(M1)  $t \iota(\lambda) t^{-1} \in \iota(\Lambda)$  for every  $\lambda \in \Lambda$ ;

(M2)  $c \iota(\lambda) = \iota(\lambda) c$  for every  $\lambda \in \Lambda$ .

Write  $d := tct^{-1}$ . The *defect subgroup* of the datum is the normal closure

$$N_{\text{comp}} = \langle\langle [d, \iota(\lambda)] : \lambda \in \Lambda \rangle\rangle_H \trianglelefteq H.$$

Both conditions are relations among generators, so a datum can be read off a presentation. The relations force  $c$  to commute with  $\iota(\Lambda)$  and force  $d$  to commute with the compressed copy  $t\iota(\Lambda)t^{-1}$ , but they do not constrain the commutators  $[d, \iota(\lambda)]$ ; the defect subgroup  $N_{\text{comp}}$  — the *compression defect* of the datum — is their normal closure. The theorems of this section show that every finite normal subgroup of  $H$  contained in  $N_{\text{comp}}$ , and more generally every normal property-(T) subgroup contained in it, lies in the kernel of every corona representation of  $H$ .

**Definition S7.2** (Hilbert–Schmidt invisibility). An element  $g$  of a countable group  $H$  is *Hilbert–Schmidt invisible* if, for every operator-norm asymptotic representation of  $H$  and every ultrafilter refining the cofinite filter, the induced homomorphism into the tracial ultraproduct of the matrix blocks maps  $g$  to the identity.

**Theorem S7.3** (finite-normal obstruction criterion). *Let  $H$  be a countable group with a Kazhdan conjugation datum  $(\Lambda, \iota, t, c)$ , and let  $F \trianglelefteq H$  be a finite normal subgroup with  $F \subseteq N_{\text{comp}}$ . Then every corona representation  $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$  satisfies  $\Theta(f) = 1$  for all  $f \in F$ .*

*Proof. Tracial vanishing of the defect.* Let  $(U_{g,n})$  be coordinate unitary lifts of any operator-norm almost representation of  $H$  (in particular, of a corona representation), and put

$$K_2 = \{g \in H : \|U_{g,n} - 1\|_2 \longrightarrow 0\}.$$

Unitary invariance and asymptotic multiplicativity make  $K_2 \trianglelefteq H$ . Since  $c$  centralizes  $\iota(\Lambda)$ , its lifts belong to the Hilbert–Schmidt asymptotic commutant of  $\iota(\Lambda)$ . Applying Theorem 3.1 to  $x_n = U_{c,n}$  puts the lifts of  $d = tct^{-1}$  there as well. Therefore  $[d, \iota(\lambda)] \in K_2$  for every  $\lambda \in \Lambda$ ; since  $K_2$  is normal, it contains the normal closure of these commutators:  $N_{\text{comp}} \subseteq K_2$ , and so  $F \subseteq K_2$ . Convergence of the full sequence gives triviality along every ultrafilter refining the cofinite filter, so every element of  $N_{\text{comp}}$  is Hilbert–Schmidt invisible (Definition S7.2).

*The coordinate Reynolds corner.* Suppose a corona representation  $\Theta$  is nontrivial on  $F$ ; choose coordinate unitary lifts  $(V_{g,n})$  of  $\Theta$  and form the coordinate Reynolds averages

$$p_n = \frac{1}{|F|} \sum_{f \in F} V_{f,n}.$$

Because the lifts are asymptotically multiplicative and  $F$  is finite,  $(p_n)$  is asymptotically self-adjoint and idempotent, and normality of  $F$  makes it asymptotically central for all of  $\Theta(H)$ . The Hermitian part of  $p_n$  is asymptotically idempotent, so its spectrum clusters at  $\{0, 1\}$  and rounding at  $1/2$  by continuous functional calculus yields genuine projections at operator-norm distance  $o(1)$  from  $p_n$ ; the rounded projections inherit the asymptotic centrality. Let  $q_n$  be their complements. Nontriviality of  $\Theta$  on  $F$  forces  $q_n \neq 0$  along infinitely many coordinates — otherwise the averages converge to 1 and  $\Theta|_F$  is trivial. Retain exactly those coordinates and relabel them by  $\mathbb{N}$ . Compress every lift to the corner  $q_n M_{d_n}(\mathbb{C}) q_n \cong M_{r_n}(\mathbb{C})$  with  $r_n = \text{rank } q_n > 0$ , and polar-correct the compressed matrices: asymptotic centrality makes each compression  $q_n V_{g,n} q_n$  almost unitary in the corner, so its polar unitary is at operator-norm distance  $o(1)$  and almost-multiplicativity survives the correction. The result is an operator-norm almost representation  $(W_{g,n})$  of  $H$  on the nonzero corner blocks. Since the corner is the complement of the rounded average, the finite average vanishes there:

$$(*) \quad \left\| \sum_{f \in F} W_{f,n} \right\| \longrightarrow 0 \quad \text{in operator norm.}$$

*The contradiction.* The tracial vanishing applies to the corner almost representation  $(W_{g,n})$  itself: every  $f \in F$  satisfies  $\|W_{f,n} - 1\|_2 \rightarrow 0$ . Since  $F$  is finite, the maximum of these  $|F|$  null sequences is itself null, so

$$\sum_{f \in F} W_{f,n} \longrightarrow |F|1 \quad \text{in normalized Hilbert–Schmidt norm,}$$

each block carrying its own normalized trace. This contradicts  $(*)$ , because operator-norm convergence dominates normalized Hilbert–Schmidt convergence. Hence no corona representation is nontrivial on  $F$ .  $\square$

The Reynolds-corner argument used only one property of  $N_{\text{comp}}$ : every element is Hilbert–Schmidt invisible in every operator-norm almost representation of  $H$ . Any normal subgroup with that tracial-vanishing property therefore obstructs its finite normal subgroups in exactly the same way; Theorem S7.5 exploits this.

The central-sign criterion of the main text (Theorem 4.1) is the case  $F = \{1, \varepsilon\}$  of Theorem S7.3: there the Reynolds average is  $p_n = \frac{1}{2}(1 + V_{\varepsilon,n})$ , its complementary corner is the negative spectral corner of the sign, and compression to that corner gives a model in which  $\varepsilon = -1$  exactly. Nothing else in the two-element case is special, and the general statement carries no centrality hypothesis.

**S7.1 The intrinsic defect** The transport estimate applies to every one-sided conjugator of a subgroup, not only to the element  $t$ . For a subgroup  $L \leq H$ , let  $G_{\text{comp}}(L)$  be generated by the elements  $s$  satisfying  $sLs^{-1} \subseteq L$  — the *one-sided conjugators* of  $L$  — and put

$$(9) \quad \mathfrak{D}(H, L) = \left\langle\left\langle [gzg^{-1}, \ell] : g \in G_{\text{comp}}(L), z \in C_H(L), \ell \in L \right\rangle\right\rangle_H,$$

the *intrinsic defect* of  $L$  in  $H$ . It depends only on the pair  $(H, L)$ , not on a choice of datum.

**Corollary S7.4** (transport along all one-sided conjugators). *Let  $(r_n)$  and  $V_{g,n} \in \mathcal{U}(r_n)$  be an operator-norm asymptotic representation of  $H$ , let  $\Lambda$  have property (T), let  $\iota : \Lambda \rightarrow H$  be a homomorphism, and put  $L = \iota(\Lambda)$ . Suppose that  $(x_n \in M_{r_n}(\mathbb{C}))$  is uniformly bounded in operator norm and belongs to the Hilbert–Schmidt asymptotic commutant of  $L$ :*

$$\left\| x_n - V_{\iota(\lambda),n} x_n V_{\iota(\lambda),n}^* \right\|_2 \longrightarrow 0 \quad (\lambda \in \Lambda).$$

*Then every  $g \in G_{\text{comp}}(L)$  acts in both directions on that asymptotic commutant. Explicitly, both*

$$V_{g,n} x_n V_{g,n}^* \quad \text{and} \quad V_{g,n}^* x_n V_{g,n}$$

*satisfy the preceding convergence for every  $\lambda \in \Lambda$ .*

*Proof.* For a one-sided conjugator  $s$ , Theorem 3.1 gives forward transport. Reverse transport follows from the same proof: the equality  $P = V P V^*$  obtained there gives  $V^* P V = P$  as well, so the fixed subspace is preserved by conjugation in both directions. In the finite-stage proof of Section S5, the equal-rank reversal estimate reverses the one-sided containment of the spectral subspaces; the same argument then runs with the adjoint unitary, with the displacement and concentration steps unchanged. The elements acting in both directions form a subgroup of  $H$ , and hence contain the subgroup generated by all one-sided conjugators, namely  $G_{\text{comp}}(L)$ .  $\square$



**Theorem S7.5** (the intrinsic defect and the MF radical). *Let  $H$  and  $\Gamma$  be countable groups, let  $\Gamma$  have property (T), let  $\iota: \Gamma \rightarrow H$  be a homomorphism, and put  $L = \iota(\Gamma)$ . If  $F \trianglelefteq H$  is finite and  $F \leq \mathfrak{D}(H, L)$ , then every homomorphism from  $H$  to every norm matrix corona maps every element of  $F$  to the identity.*

*Proof.* Let  $(U_{h,n})$  be coordinate lifts of an operator-norm almost representation of  $H$  and put  $K_2 = \{h : \|U_{h,n} - 1\|_2 \rightarrow 0\}$ , a normal subgroup of  $H$ . If  $z \in C_H(L)$ , then  $(U_{z,n})$  lies in the asymptotic commutant of  $L$ , and Corollary S7.4 places  $(U_{gzg^{-1},n})$  there for every  $g \in G_{\text{comp}}(L)$ . Hence each generator  $[gzg^{-1}, \ell]$ ,  $\ell \in L$ , displayed in (9) belongs to  $K_2$ ; since  $K_2 \trianglelefteq H$ , it contains their normal closure:  $\mathfrak{D}(H, L) \leq K_2$ . Every element of  $\mathfrak{D}(H, L)$  is therefore Hilbert–Schmidt invisible in every operator-norm almost representation of  $H$ , and the coordinate Reynolds corner of Theorem S7.3 applies verbatim to any finite normal  $F \leq \mathfrak{D}(H, L)$ .  $\square$

**S7.2 The normal-Kazhdan form of the obstruction** Finiteness of  $F$  enters Theorem S7.3 only through the Reynolds corner. A finite group has property (T), and its Reynolds average is its fixed-space average; for an arbitrary normal property-(T) subgroup, the spectral projection of a Kazhdan average takes its place. The resulting form of the obstruction has no finiteness, centrality, or torsion hypothesis; it is used in Section S12.

**Theorem S7.6** (normal-Kazhdan obstruction). *Let  $H$  be a countable group with a Kazhdan conjugation datum  $(\Lambda, \iota, t, c)$ , and let  $K \trianglelefteq H$  be a normal subgroup with property (T) such that  $K \subseteq N_{\text{comp}}$ . Then every corona representation  $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$  satisfies  $\Theta(k) = 1$  for all  $k \in K$ .*

Taking  $K = F$  finite recovers Theorem S7.3, since finite groups have property (T): the whole group with tolerance 1 is a Kazhdan pair, by orbit averaging. In Theorem S7.6,  $K$  may be infinite, noncentral, and torsion-free.

*Proof.* The containment  $N_{\text{comp}} \subseteq K_2$  from the proof of Theorem S7.3 uses no finiteness, so every element of  $K$  is Hilbert–Schmidt invisible in every operator-norm almost representation of  $H$ . It remains to replace the Reynolds corner by a corner adapted to  $K$ ; the spectral projection of a Kazhdan average provides it.

Suppose a corona representation  $\Theta$  is nontrivial on  $K$ . Its image  $\bar{H} = \Theta(H)$  is countable, hence MF, and Proposition 1.1, applied along an exhaustion of  $\bar{H}$ , gives an operator-norm asymptotic representation  $(V_{g,n})$  in which every fixed nontrivial element eventually stays at operator-norm distance at least 1 from the identity. These models bound only operator-norm distances; at growing dimension they give no lower bound on normalized Hilbert–Schmidt distances, and the corner constructed below provides that lower bound. The datum and the subgroup pass to their  $\Theta$ -images with normality and property (T) preserved, and  $\bar{K} = \Theta(K)$  is nontrivial and lies in the defect subgroup of the image datum; it suffices to derive a contradiction for  $\bar{H}$ .



*The moving spectral subspace.* Fix a finite symmetric generating Kazhdan set  $S \subseteq \bar{K}$  with constant  $\varepsilon_0$ , at each coordinate let  $h_n$  be the Hermitian part of the average of the models of  $S$ , and cut the spectrum of  $h_n$  at  $\theta = 1 - \varepsilon_0^2/(4|S|)$ . Let  $q_n$  be the spectral projection of  $h_n$  for  $[-1, \theta]$ : the *moving spectral subspace*. On its complement — the top spectral subspace — every model of every element of  $\bar{K}$  is close to the identity in operator norm, by the Kazhdan-pair displacement estimate. If  $q_n = 0$  for all large  $n$ , every element of  $\bar{K}$  would satisfy  $\|V_{k,n} - 1\| \rightarrow 0$ , contradicting the operator-norm separation of any fixed nontrivial element of  $\bar{K}$ . So the moving subspace is nonzero along an infinite set of coordinates, which we retain.

*The tracial deficit on the corner.* Equip each retained corner block  $q_n M_{d_n}(\mathbb{C}) q_n$  with its renormalized trace. By the definition of the cut,  $q_n h_n q_n \leq \theta q_n$ , so the average over  $S$  of the real parts of the renormalized traces of the compressed models is at most  $\theta$ . Hence at every retained coordinate some generator in  $S$  has compressed model with real renormalized trace at most  $\theta$ , and therefore normalized Hilbert–Schmidt distance at least  $\sqrt{2(1-\theta)} - o(1)$  from the corner identity after polar correction; passing to a subsequence fixes a single such generator  $s_0$ .

*Compression.* Normality makes the cut asymptotically central. Conjugating by an ambient model  $V_{g,n}$  carries  $h_n$  to the Hermitian average over the conjugate generating set  $gSg^{-1}$  of  $\bar{K}$ , which has the same Kazhdan constant. The two estimates of Section S5 now apply to either average symmetrically: a vector in the top spectral subspace of one average is almost fixed by each generator of the other set (displacement, transported across the finitely many conjugations by almost-multiplicativity), hence almost lies in the top spectral subspace of the other average (the concentration inequality). Two-sided almost-containment of the two subspaces forces  $\|q_n - V_{g,n} q_n V_{g,n}^*\| \rightarrow 0$ , that is,  $\|[q_n, V_{g,n}]\| \rightarrow 0$  for every  $g \in \bar{H}$ . Compressing every model to the moving corner and polar-correcting yields an operator-norm asymptotic representation of all of  $\bar{H}$  on nonzero blocks, in which  $s_0$  keeps a definite normalized Hilbert–Schmidt distance from the identity.

*The contradiction.* The tracial vanishing applies verbatim to the corner representation:  $\bar{K}$  lies in the defect subgroup of the image datum, so every element of  $\bar{K}$  — in particular  $s_0$  — is Hilbert–Schmidt invisible on the corner, contradicting the tracial deficit of  $s_0$ . The coordinates are scheduled as in the proof of Theorem S7.3.  $\square$

The proofs above use the conjugation datum through one property only: every element of the defect subgroup is Hilbert–Schmidt invisible in every operator-norm almost representation. The next theorem takes that invisibility as its hypothesis.

**Theorem S7.7** (abstract invisibility obstruction). *Let  $H$  be a countable group and let  $D \leq H$  be a subgroup every element of which is Hilbert–Schmidt invisible (Definition S7.2). Let  $K \trianglelefteq H$  be a normal subgroup with property (T)*

such that  $K \leq D$ . Then every corona representation  $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$  satisfies  $\Theta(k) = 1$  for all  $k \in K$ .

*Proof.* The proof of Theorem S7.6 used its datum only to make every element of the defect subgroup Hilbert–Schmidt invisible in every operator-norm almost representation; here that invisibility is the hypothesis. Suppose a corona representation  $\Theta$  is nontrivial on  $K$ . As there, the image  $\bar{H} = \Theta(H)$  is countable and MF, and Proposition 1.1 provides an operator-norm asymptotic representation of  $\bar{H}$  that separates every fixed nontrivial element. Every operator-norm almost representation of  $\bar{H}$  composes with  $\Theta$  to one of  $H$ , so every element of  $\Theta(D)$  inherits the invisibility hypothesis, while  $\bar{K} = \Theta(K) \leq \Theta(D)$  is normal, Kazhdan, and nontrivial. The moving spectral subspace, the tracial deficit, and the compression step of the proof of Theorem S7.6 apply verbatim to  $\bar{K}$ , and the contradiction step uses the inherited invisibility in place of the defect containment.  $\square$

Applied to the intrinsic defect, Theorem S7.7 gives the normal-Kazhdan companion of Theorem S7.5:

**Corollary S7.8** (normal-Kazhdan form of the intrinsic radical). *Let  $H$  and  $\Gamma$  be countable groups, let  $\Gamma$  have property (T), let  $\iota: \Gamma \rightarrow H$  be a homomorphism, and put  $L = \iota(\Gamma)$ . If  $K \trianglelefteq H$  is a normal subgroup with property (T) and  $K \leq \mathfrak{D}(H, L)$ , then every homomorphism from  $H$  to every norm matrix corona maps every element of  $K$  to the identity.*

*Proof.* The proof of Theorem S7.5 shows that every element of  $\mathfrak{D}(H, L)$  is Hilbert–Schmidt invisible in every operator-norm almost representation of  $H$ . Theorem S7.7 with  $D = \mathfrak{D}(H, L)$  concludes.  $\square$

The subgroup generated by *all* normal property-(T) subgroups of  $\mathfrak{D}(H, L)$  therefore also lies in the kernel of every corona representation.

## S8 THE FINITE-DIMENSIONAL OBSTRUCTION

The finite-dimensional analogue of the obstruction is elementary: it needs finite dimensionality and the one-sided conjugation and centralizer relations alone, and its conclusion covers the entire intrinsic defect (9):

**Theorem S8.1** (the exact finite-dimensional analogue). *Let  $k$  be a field,  $V$  a finite-dimensional  $k$ -vector space,  $H$  a group,  $\Gamma \leq H$  a subgroup, and  $t, c \in H$ ,  $a \in \Gamma$  elements with  $t\Gamma t^{-1} \subseteq \Gamma$  and  $c\gamma = \gamma c$  for all  $\gamma \in \Gamma$ . Then every homomorphism  $\pi: H \rightarrow \text{GL}(V)$  satisfies*

$$\pi([tct^{-1}, a(tct^{-1})a^{-1}]) = 1.$$

*More generally, every such  $\pi$  sends the entire subgroup  $\mathfrak{D}(H, \Gamma)$  to the identity.*

Applied to  $H = E$ , Theorem S8.1 shows that *every* finite-dimensional representation of  $E$  over *every* field sends  $w$  to the identity. Consequently finite-dimensional linear representations do not separate the points of  $E$ ,

no finite quotient can witness  $w \neq 1$ , and the infinite Clifford witness of Section S3 cannot be replaced by a finite-dimensional one. In the terminology of [BDL, Definition 1.2], the central involution  $w$ , invisible in every finite quotient, makes  $E$  a group of *Deligne type*. Such groups were proposed there as the natural candidate counterexamples for all three remaining approximation problems, and  $E$  realizes that proposal at the operator-norm endpoint. Property (T) enters the corona argument through the Kazhdan fixed-space projection of the adjoint actions — in quantitative form, a finite-stage spectral subspace at each coordinate, where the equal-rank reversal estimate of Section S5 plays the role that equal dimension plays in the exact argument below.

*Proof of Theorem S8.1.* Let  $\pi: H \rightarrow \mathrm{GL}(V)$  be a homomorphism and let

$$C = \{x \in \mathrm{End}_k(V) : \pi(\gamma)x = x\pi(\gamma) \text{ for all } \gamma \in \Gamma\}$$

be the commutant of  $\pi(\Gamma)$ , a linear subspace of  $\mathrm{End}_k(V)$ . Let  $u = \pi(t)$  and consider the linear automorphism  $\Phi(x) = uxu^{-1}$  of  $\mathrm{End}_k(V)$ .

We claim  $C \subseteq \Phi(C)$ . Let  $x \in C$  and set  $y = u^{-1}xu$ , so that  $x = \Phi(y)$ . For  $\gamma \in \Gamma$  we have  $t\gamma t^{-1} \in \Gamma$ , hence  $\pi(t\gamma t^{-1})$  commutes with  $x$ , and therefore

$$\pi(\gamma)y = u^{-1}\pi(t\gamma t^{-1})xu = u^{-1}x\pi(t\gamma t^{-1})u = y\pi(\gamma).$$

Thus  $y \in C$ , proving the claim. Since  $\Phi$  is a linear isomorphism,  $\dim_k \Phi(C) = \dim_k C < \infty$ , and the inclusion  $C \subseteq \Phi(C)$  forces

$$(10) \quad C = \Phi(C) = uCu^{-1}.$$

Now  $\pi(c) \in C$  because  $c$  centralizes  $\Gamma$ . By (10),  $\pi(tct^{-1}) = u\pi(c)u^{-1} \in C$ . Since  $a \in \Gamma$ , the element  $\pi(tct^{-1})$  commutes with  $\pi(a)$ , so  $\pi(a(tct^{-1})a^{-1}) = \pi(tct^{-1})$ , and therefore

$$\pi([tct^{-1}, a(tct^{-1})a^{-1}]) = [\pi(tct^{-1}), \pi(tct^{-1})] = 1. \quad \square$$

The same calculation proves the final, subgroup-valued assertion of Theorem S8.1. The elements of  $H$  whose conjugation action preserves  $C$  in both directions form a subgroup. The argument above puts every one-sided conjugator in that subgroup, and therefore puts all of  $G_{\mathrm{comp}}(\Gamma)$  in it. Thus, for  $g \in G_{\mathrm{comp}}(\Gamma)$  and  $z \in C_H(\Gamma)$ , the operator  $\pi(gzg^{-1})$  still belongs to  $C$  and commutes with every  $\pi(\ell)$ ,  $\ell \in \Gamma$ . Hence every displayed generator in (9) lies in  $\ker \pi$ ; since  $\ker \pi \trianglelefteq H$ , the whole normal closure  $\mathfrak{D}(H, \Gamma)$  does as well.

**Corollary S8.2.** *The group  $E$  of Definition S2.1 admits no injective homomorphism into  $\mathrm{GL}(V)$  for any finite-dimensional vector space  $V$  over any field. More strongly, every finite-dimensional linear representation maps  $w$  to the identity, and every homomorphism to a finite group maps  $w$  to the identity.*

*Proof.* Apply Theorem S8.1 with  $H = E$ , the image of  $\mathcal{B}$ , and the elements  $t, c, v_1$ ; the hypotheses hold by the defining relations (8). Every finite-dimensional representation kills  $w$ , and  $w \neq 1$  by Proposition S3.3. For a

finite target, compose with its faithful left regular representation on the group algebra over  $\mathbb{Q}$ . The composite is a finite-dimensional linear representation, so it kills  $w$ . Evaluating the resulting identity matrix at the basis vector of the identity element returns the image of  $w$  in the finite group.  $\square$

*Remark S8.3* (the invariant-size principle). The proof of Theorem S8.1 used exactly one feature of finite dimensionality: a subspace of a finite-dimensional space that contains an isomorphic copy of itself equals it. In abstract form: suppose a conjugation-invariant size  $\sigma$  with values in  $\mathbb{N}$  is defined on the subgroups of a group  $Q$  and satisfies the separation property that  $K \leq L$  with  $\sigma(L) \leq \sigma(K)$  forces  $K = L$ . If  $gKg^{-1} \subseteq K$ , then  $gKg^{-1}$  and  $K$  have equal size, so the inclusion is an equality: such a target admits no strict one-sided compression. Cardinality on the subgroups of a finite group qualifies, so no finite quotient detects strict one-sided compression. The finite-dimensional case is not the same principle applied to subgroups. The natural candidate size there, the dimension of the fixed space of the adjoint action, is  $\dim_k \text{End}_H(V)$ , the dimension of the commutant; it is order-reversing rather than order-preserving in  $H$ , and it does not separate nested subgroups —  $A_4 < S_4$  in the standard three-dimensional representation have the same commutant  $k \cdot 1$ . What the proof of Theorem S8.1 uses is the principle one level down, on the commutant itself: the inclusion  $C \subseteq uCu^{-1}$  between subspaces of equal finite dimension is an equality (10). Pinning the commutant, not the subgroup, is what kills the defect. The corona argument of Theorem A is the analytic counterpart of this principle at infinite dimension.

*Remark S8.4* (chain conditions suffice). The separation property of the size can be dropped, at the price of concluding equality of values rather than of subgroups. An order isomorphism of a poset with no infinite strictly ascending chain fixes every point that it moves weakly upward, and dually with descending chains; consequently a conjugation-equivariant monotone invariant with chain-condition values takes the same value on  $\Gamma$  and on  $t\Gamma t^{-1}$ . In particular, for every finite-dimensional representation  $\sigma$  of  $H$  over every field, the fixed subspaces of the subgroup and of its one-sided conjugate coincide. Applied to the representations induced on tensor constructions —  $V^{\otimes p} \otimes (V^*)^{\otimes q}$  with the contragredient action, finite direct sums, Hom-spaces — compression creates no invariant tensors of any finite type; the commutant identity (10) is the case  $\text{End}(V) \cong V \otimes V^*$ . Taking the invariant to be the Zariski closure of the image in  $\text{GL}(V)$ , whose closed subgroups satisfy the descending chain condition, gives  $\overline{\sigma(t\Gamma t^{-1})}^Z = \overline{\sigma(\Gamma)}^Z$ : an exact finite-dimensional representation assigns the compressed subgroup and the subgroup the same algebraic envelope. Only the chain condition is used: whenever the closed subgroups are well-founded under  $\supseteq$  and the envelope is monotone and conjugation-stable, a one-sided compression leaves the envelope fixed. The chain condition for Zariski-closed subgroups of  $\text{GL}(V)$  is classical and is quoted here. Theorem S6.1(2) is the asymptotic counterpart at fixed tensor type.

Why the corona argument is different. An MF model is only asymptotically multiplicative, so its coordinate fixed spaces are not literal invariant subspaces. Operator-norm control keeps the coordinate adjoint actions almost multiplicative on the Hilbert–Schmidt spaces. Property (T) then supplies an almost-fixed spectral subspace. The equal-rank reversal estimate of Section S5 turns its one-sided compression into equality. These are the quantitative counterparts of the exact finite-dimensional steps above.

## S9 THE CYCLIC COMPARISON

The finite-dimensional calculation of Section S8 does not by itself determine whether its marked word is trivial:

**Theorem S9.1** (the cyclic comparison). *Define the marked presentation*

$$E_{\text{BS}} = \left\langle \gamma_0, t, c \left| \begin{array}{l} t\gamma_0 t^{-1} = \gamma_0^2, \quad c^2 = 1, \quad [c, \gamma_0] = 1, \\ w_{\text{BS}}^2 = 1, \quad [w_{\text{BS}}, \gamma_0] = [w_{\text{BS}}, t] = [w_{\text{BS}}, c] = 1 \end{array} \right. \right\rangle,$$

where  $w_{\text{BS}} = [tct^{-1}, \gamma_0(tct^{-1})\gamma_0^{-1}]$ . Then  $E_{\text{BS}}$  is finitely presented and  $w_{\text{BS}} \neq 1$ . There is an explicit surjective homomorphism from  $E_{\text{BS}}$  onto its realized Clifford quotient that maps  $w_{\text{BS}}$  to the nontrivial central sign. Nevertheless every finite-dimensional representation of  $E_{\text{BS}}$  over every field maps  $w_{\text{BS}}$  to the identity.

Thus exact finite-dimensional invisibility does not by itself force triviality of the marked word. The MF conclusion in Theorem A is the separate uniform assertion supplied by the fixed-space argument.

*Proof of Theorem S9.1.* As with  $w$ , the relator  $w_{\text{BS}}^2 = 1$  is redundant:  $tct^{-1}$  and  $\gamma_0(tct^{-1})\gamma_0^{-1}$  are involutions, so centrality forces it by (4). It is retained only to display the marked pattern. Deleting it gives a six-relator presentation of the same group. The stable relation gives  $t\langle\gamma_0\rangle t^{-1} \subseteq \langle\gamma_0\rangle$ , and  $c$  centralizes this cyclic base. The same commutant argument as in Theorem S8.1 therefore maps  $w_{\text{BS}}$  to the identity in every finite-dimensional representation over every field.

For nontriviality, use the affine realization of  $\text{BS}(1, 2)$  and its coset action. The root site is fixed by the cyclic base, while the two sites represented by  $t$  and  $\gamma_0 t$  are distinct. Attach Clifford generators to these sites and map  $c$  to the root lamp. All seven relators hold, and

$$w_{\text{BS}} \mapsto [c_{t\langle\gamma_0\rangle}, c_{\gamma_0 t\langle\gamma_0\rangle}] = -1.$$

Thus  $w_{\text{BS}} \neq 1$ . Restricting the target to the subgroup generated by the three displayed images gives the asserted surjective realized quotient.  $\square$

Sharpness of the Kazhdan hypothesis. The realized Clifford quotient in Theorem S9.1 is amenable, hence MF, and its quotient map detects  $w_{\text{BS}}$ . Thus some corona representation of  $E_{\text{BS}}$  separates the marked word from the identity, and property (T) cannot be omitted from the central-sign criterion. In the radical language of Section S11,  $w_{\text{BS}} \notin \text{Rad}_{\text{MF}}(E_{\text{BS}})$ . The affine group

$BS(1, 2) \cong \mathbb{Z}[1/2] \rtimes \mathbb{Z}$  is solvable. The Clifford lamp group on its coset space is locally finite, so their semidirect product is amenable. The realized quotient is a subgroup of this amenable group and is therefore amenable, hence MF by [TWW]. Its marked word is the nontrivial Clifford sign by the calculation above.

This detecting corona representation is necessarily approximate: neither the group  $E_{BS}$  nor its realized Clifford quotient is LEF, and the latter is not residually finite. In every finite group the doubling relation makes the image of  $\langle \gamma_0^2 \rangle$  equal to the image of  $\langle \gamma_0 \rangle$ ; the lamp therefore commutes with the required “halved” element and the marked word maps to the identity, so neither group is residually finite. The same equations transport through any exact local embedding of a finite test set containing the defining relators and the marked word, so an exact finite local model would already force the marked word to the identity: neither group is LEF.

## Part III. Structure of the example and its consequences

### S10 THE SCALING FAMILY

The same construction works for every exponent. For  $m \geq 2$ , let  $E_m$  be the same eight-generator presentation, replacing the three relations  $tv_it^{-1} = v_i^2$  by  $tv_it^{-1} = v_i^m$  and leaving every other relator unchanged; thus  $E_2 = E$ . Write  $w_m$  for the image of the same marked word  $w$ .

**Corollary S10.1** (the scaling family). *For every natural number  $m \geq 2$ , the group  $E_m$  is finitely presented and  $w_m$  is a nontrivial central involution mapped to the identity by every homomorphism from  $E_m$  to every unitary norm matrix corona. Consequently  $E_m$  is not MF, and neither  $C_{\max}^*(E_m)$  nor  $C_{\text{red}}^*(E_m)$  is an MF  $C^*$ -algebra.*

*Proof.* Replace the affine dilation  $\text{diag}(2, 2, 2, 1)$  by  $\text{diag}(m, m, m, 1)$  (both acting as in Lemma S3.2). It conjugates each integral translation  $v_i$  to  $v_i^m$  and fixes the three linear generators. Its inverse sends  $v_1$  to translation by  $e_1/m$ , which is not integral; hence the two Clifford sites remain distinct and  $w_m$  maps to  $-1$ . The Kazhdan base, the one-sided conjugation argument, and the central-sign criterion are independent of  $m$ .  $\square$

### S11 CONSEQUENCES

This section develops the universal MF radical and the basic permanence facts, computes radicals by orbit collapse, constructs a sofic non-MF group, and proves the reduced-group- $C^*$  consequence. They all come from the same marked element.

#### S11.1 The MF radical

**Definition S11.1.** For a countable group  $G_1$ , the *MF radical* is

$$\text{Rad}_{\text{MF}}(G_1) = \bigcap \{ \ker \Theta : \Theta \text{ a corona representation of } G_1 \},$$

a normal subgroup of  $G_1$ . The intersection does not change if the dimension sequences are allowed to be arbitrary positive rather than strictly increasing (Proposition 1.1). This is the operator-norm analogue of the invisible subgroups of [Slo17, Definitions 2.5–2.7] and of the metric-ultraproduct approximation framework of [NST].

**Lemma S11.2** (functoriality). *For every homomorphism  $\varphi: G_1 \rightarrow G_2$  of countable groups,*

$$\varphi(\text{Rad}_{\text{MF}}(G_1)) \subseteq \text{Rad}_{\text{MF}}(G_2) :$$

*the MF radical is functorial. In particular, if  $\varphi: E \rightarrow G_2$  satisfies  $\varphi(w) \neq 1$ , then  $G_2$  is not MF.*

*Proof.* If  $x \in \text{Rad}_{\text{MF}}(G_1)$  and  $\Theta$  is a corona representation of  $G_2$ , then  $\Theta \circ \varphi$  is a corona representation of  $G_1$ , so  $\Theta(\varphi(x)) = 1$ . For the second statement,  $w \in \text{Rad}_{\text{MF}}(E)$  by Theorem A, so  $1 \neq \varphi(w) \in \text{Rad}_{\text{MF}}(G_2)$ , and an injective corona representation of  $G_2$  would kill  $\varphi(w)$ , a contradiction.  $\square$

**Proposition S11.3** (the universal MF quotient). *Let  $G_1$  be a countable group and  $R = \text{Rad}_{\text{MF}}(G_1)$ .*

- (1)  *$R$  is the kernel of a single corona representation of  $G_1$ .*
- (2)  *$G_1/R$  is MF.*
- (3) *Every homomorphism from  $G_1$  to an MF group factors through  $G_1/R$ .*

*In particular  $G_1$  is MF if and only if  $R = \{1\}$ .*

*Proof.* (3) If  $\varphi: G_1 \rightarrow M$  with  $M$  MF and  $j: M \rightarrow \text{U}_{\text{cor}}((d_n))$  is an injective corona representation, then  $j \circ \varphi$  is a corona representation of  $G_1$ , so  $j(\varphi(R)) = 1$  and, by injectivity of  $j$ ,  $\varphi(R) = 1$ .

(1) and (2). Work on  $Q = G_1/R$ . Every corona representation of  $G_1$  kills  $R$ , and each  $g \notin R$  is detected by one, which descends; so every nonidentity element of  $Q$  is detected by a corona representation of  $Q$ . Enumerate the nonidentity elements and their detectors, pass to coordinate unitary lifts, and diagonalize: at stage  $k$ , select in each of the first  $k$  detectors a coordinate at which multiplicativity holds within  $1/k$  on the first  $k$  group elements and each of the first  $k$  marked elements stays separated by its fixed positive constant — possible because each detector separates its element along infinitely many coordinates — and take the block sum. Block operator norms are maxima, so the resulting sequence is an almost representation detecting every nonidentity element; its class, composed with the polar-correction isomorphism of Lemma 2.2, is a faithful corona representation of  $Q$ . This proves (2), and composing with  $G_1 \twoheadrightarrow Q$  proves (1). Finally, if  $G_1$  is MF then an injective corona representation forces  $R = \{1\}$ ; if  $R = \{1\}$  then (2) says  $G_1$  is MF.  $\square$

**Corollary S11.4** (exact radical from a candidate quotient). *Let  $N \trianglelefteq G_1$  satisfy  $N \leq \text{Rad}_{\text{MF}}(G_1)$ . Every corona representation of  $G_1$  factors uniquely through  $G_1/N$ . If in addition  $G_1/N$  is MF, then*

$$\text{Rad}_{\text{MF}}(G_1) = N.$$

*Proof.* The containment  $N \leq \ker \Theta$  gives the unique quotient factorization for every corona representation  $\Theta$ . For the reverse containment in the displayed equality, compose the quotient map  $G_1 \rightarrow G_1/N$  with an injective corona representation of  $G_1/N$ : its kernel is exactly  $N$ , while the MF radical lies in the kernel of every corona representation.  $\square$

The MF hypothesis on the quotient can be dropped entirely; what remains is an unconditional reduction.



**Corollary S11.5** (radical reduction to the quotient). *Let  $N \trianglelefteq G_1$  satisfy  $N \leq \text{Rad}_{\text{MF}}(G_1)$  and let  $q: G_1 \rightarrow G_1/N$  be the quotient map. Then*

$$\text{Rad}_{\text{MF}}(G_1) = q^{-1}(\text{Rad}_{\text{MF}}(G_1/N)).$$

*Proof.* For the forward containment, Lemma S11.2 gives  $q(\text{Rad}_{\text{MF}}(G_1)) \subseteq \text{Rad}_{\text{MF}}(G_1/N)$ . Conversely, suppose  $q(x) \in \text{Rad}_{\text{MF}}(G_1/N)$  and let  $\Theta$  be a corona representation of  $G_1$ . Every corona representation of  $G_1$  kills  $N$  by hypothesis, so  $\Theta$  factors as  $\bar{\Theta} \circ q$  with  $\bar{\Theta}$  a corona representation of  $G_1/N$ ; hence  $\Theta(x) = \bar{\Theta}(q(x)) = 1$ .  $\square$

Corollary S11.4 is the case of trivial quotient radical. Corollary S11.5 needs no hypothesis on the quotient: the corona representation theory of  $G_1$  is that of  $G_1/N$ , whether or not the latter is MF.

By Corollary S11.5, the passage from  $N$  to the full preimage of the quotient radical is a closure operation on the normal subgroups of a fixed group.

**Proposition S11.6** (the MF closure of relations). *For a countable group  $G_1$  and  $N \trianglelefteq G_1$  with quotient map  $q: G_1 \rightarrow G_1/N$ , put*

$$\text{cl}_{\text{MF}}(N) = q^{-1}(\text{Rad}_{\text{MF}}(G_1/N)).$$

*An element belongs to  $\text{cl}_{\text{MF}}(N)$  exactly when every corona representation of  $G_1$  that maps  $N$  to the identity maps it to the identity. The operation  $N \mapsto \text{cl}_{\text{MF}}(N)$  is extensive, monotone, and idempotent on normal subgroups;  $G_1/\text{cl}_{\text{MF}}(N)$  is MF; and  $\text{cl}_{\text{MF}}(N) = N$  exactly when  $G_1/N$  is MF.*

*Proof.* Corona representations of  $G_1$  killing  $N$  correspond to corona representations of  $G_1/N$ , which gives the membership characterization; extensivity, monotonicity, and idempotence are read off from it. The quotient clause is Proposition S11.3(2) transported along the third isomorphism theorem, and the fixed-point clause is the final equivalence of Proposition S11.3.  $\square$

For a presentation  $G = \langle X \mid R \rangle = F/N$ , the closure of  $N$  consists of the relations satisfied by every solution of  $R$  in every MF group, and the group is MF exactly when  $N$  is closed. Proposition S11.22 displays one finite fragment of this closure, and the marked element of Theorem A is an element of  $\text{cl}_{\text{MF}}(N) \setminus N$  for the forty-one relators.

**Proposition S11.7** (iterated sound rules). *Consider the subgroups of  $G_1$  reachable from the trivial subgroup by two operations: from a reached normal subgroup  $N$ , pass to any subgroup of  $\text{cl}_{\text{MF}}(N)$ ; and from any family of reached subgroups, pass to their join, the subgroup generated by their union. Every subgroup so reached lies in  $\text{Rad}_{\text{MF}}(G_1)$ ; a reached normal subgroup with MF quotient equals  $\text{Rad}_{\text{MF}}(G_1)$ ; and if the whole group is reached, every homomorphism from  $G_1$  to a countable MF group is trivial.*

*Proof.* Induct over the two operations. If a reached normal subgroup  $N$  satisfies  $N \leq \text{Rad}_{\text{MF}}(G_1)$ , then Corollary S11.5 gives  $\text{cl}_{\text{MF}}(N) = \text{Rad}_{\text{MF}}(G_1)$ , so every subgroup chosen in the first operation is again inside the radical — normal or not. For the second operation, the radical is itself a subgroup,

hence an upper bound in the subgroup lattice for any family contained in it; so the join  $\bigvee_i H_i = \langle \bigcup_i H_i \rangle$  lies in the radical too. The union itself need not be a subgroup, and it is the join that the calculus uses. The final two assertions are Corollary S11.4 and Proposition S11.3; the second of them needs  $N$  normal, which is where the quotient  $G_1/N$  is formed.  $\square$

Theorems S7.3, S7.6, and S11.13 are sound rules in this sense: each places a subgroup inside the radical of every countable group carrying its data, hence inside the radical of every reached quotient. Theorem S11.19 is the case in which a single application reaches the whole group. A rule instance can hold at a later stage without holding at an earlier one: modulo the earlier stage an element may acquire finite order, an orbit may begin to commute, or a subgroup may become normal.

**Corollary S11.8.** *Let  $A$  be a unital  $C^*$ -algebra admitting an injective, possibly nonunital,  $*$ -homomorphism into a norm matrix corona. Then the group  $E$  admits no injective homomorphism into  $\mathcal{U}(A)$ .*

*Proof.* Let  $j: A \rightarrow Q$  be the possibly nonunital inclusion into the ambient corona and let  $\phi: E \rightarrow \mathcal{U}(A)$ . Complement correction gives

$$\hat{j}(u) = j(u) + 1 - j(1_A) \in \mathcal{U}(Q).$$

This is an injective homomorphism on  $\mathcal{U}(A)$ , so  $\hat{j} \circ \phi$  is a corona representation of  $E$ . By Theorem A(2),  $(\hat{j} \circ \phi)(w) = 1$ ; injectivity of  $\hat{j}$  gives  $\phi(w) = 1$ . Proposition S3.3 has  $w \neq 1$ , so  $\phi$  cannot be injective.  $\square$

## S11.2 Permanence and its failure

**Lemma S11.9** (permanence). *Let all groups be countable.*

- (1) *Every subgroup of an MF group is MF.*
- (2) *Every residually finite group is MF.*
- (3) *Every locally finite group is MF.*

*Proof.* (1) Restrict an injective corona representation. (2) For a finite test set  $F$ , a finite quotient injective on  $F$ , composed with its left regular permutation action, is an exact local model; distinct permutations differ in some column of a permutation matrix, giving operator-norm separation at least 1, and the local-to-corona direction of Proposition 1.1 concludes. (3) A finite test set lies in a finite subgroup  $K$ ; a map that is the identity on all of  $K$ , composed with the left regular permutation action of  $K$ , is an exact local model on the test set — products of test elements stay in  $K$  — separating as in (2).  $\square$

**Corollary S11.10** (quotients). *MF groups are not closed under quotients: the free group  $F_8$  is MF, and the explicit group  $E$  is its quotient but is not MF.*

*Proof.* Map the eight free generators onto the eight displayed generators of  $E$ . The resulting homomorphism is surjective because those generators define

the displayed presentation. The group  $F_8$  is residually finite, hence MF by Lemma S11.9(2). Theorem A shows that  $E$  is not MF.  $\square$

**S11.3 Matricial stability and the radical** Call a countable group  $G$  *point-norm matricially stable* if for every operator-norm almost representation  $(V_n)$  of  $G$  there are genuine finite-dimensional unitary representations  $\pi_n$  on the same models with  $\|V_n(g) - \pi_n(g)\| \rightarrow 0$  for every  $g \in G$ . Write  $\text{Rad}_{\text{fd}}(G)$  for the intersection of the kernels of all finite-dimensional unitary representations of  $G$ .

**Proposition S11.11** (stability identifies the radicals). *For every countable group,  $\text{Rad}_{\text{MF}}(G) \leq \text{Rad}_{\text{fd}}(G)$ . If  $G$  is point-norm matricially stable, then  $\text{Rad}_{\text{MF}}(G) = \text{Rad}_{\text{fd}}(G)$ ; in particular a stable group with  $\text{Rad}_{\text{fd}}(G) \neq 1$  is not MF.*

*Proof.* The first inclusion needs no stability. A finite-dimensional unitary representation  $\pi$  on  $\mathbb{C}^d$  amplifies to  $\pi^{\oplus n}$  on  $\mathbb{C}^{nd}$ ; the dimensions  $nd$  are strictly increasing, so  $(\pi^{\oplus n})_n$  is a corona representation in the sense of (2). Since  $\|\pi^{\oplus n}(x) - 1\| = \|\pi(x) - 1\|$  for every  $n$ , that sequence is null exactly when  $\pi(x) = 1$ . Hence an MF-invisible element dies in every finite-dimensional representation.

For the converse inclusion, assume  $G$  point-norm matricially stable, let  $x \in \text{Rad}_{\text{fd}}(G)$ , and suppose toward a contradiction that a corona representation  $\Theta$  has  $\Theta(x) \neq 1$ . Choose unitary coordinate lifts  $U_n(g)$  of  $\Theta(g)$ . Since  $\Theta(x) - 1 \neq 0$  in the norm quotient, after passing to a subsequence there is  $\delta > 0$  with  $\|U_n(x) - 1\| \geq \delta$  for every  $n$ . The lifts form an operator-norm almost representation, so stability supplies genuine representations  $\pi_n$  on the same blocks with  $\|U_n(g) - \pi_n(g)\| \rightarrow 0$  for each fixed  $g$ . But  $x \in \text{Rad}_{\text{fd}}(G)$  gives  $\pi_n(x) = 1$  for every  $n$ , so  $\|U_n(x) - 1\| = \|U_n(x) - \pi_n(x)\| \rightarrow 0$ , against the uniform lower bound.  $\square$

For a stable group the corona representation theory is therefore computed by finite-dimensional representation theory alone. Combined with the finite-dimensional blindness of Theorem S8.1, this is an obstruction axis with no torsion, no marked word, and no compression datum: any point-norm matricially stable group with a finite-dimensionally invisible element is non-MF.

**S11.4 Orbit collapse and defect saturation** The criteria of Section S7 bound the radical from below by torsion or rigidity already present in the defect subgroup. This subsection gives a second application of one-sided compression, under a commuting-orbit hypothesis in place of centrality: every corona representation identifies an entire conjugation orbit of an involution, so whole families of commutators — not one marked element — enter the radical, and by Corollary S11.5 the radical computation reduces to the quotient by all of them.

**Definition S11.12** (involutive compression witness). Let  $H$  be a group,  $L \leq H$  a subgroup, and  $s \in H$ . An element  $k \in H$  is an *involutive compression witness* for  $(L, s)$  if

- (W1)  $k^2 = 1$ ;
- (W2)  $s\gamma s^{-1}$  commutes with  $k$  for every  $\gamma \in L$ ;
- (W3) the conjugates  $\gamma_1 k \gamma_1^{-1}$  and  $\gamma_2 k \gamma_2^{-1}$  commute for all  $\gamma_1, \gamma_2 \in L$ .

The *involutive collapse defect* of the pair  $(L, s)$  is the normal subgroup

$$D_{\text{coll}}(L, s) = \langle\langle [\gamma, k] : \gamma \in L, k \text{ a witness for } (L, s) \rangle\rangle_H.$$

In a permutational wreath product  $K^{(X)} \rtimes G$  with  $L \leq G$  compressed by  $s$  and  $x_0 \in X$  a point fixed by  $sLs^{-1}$ , every involution of the lamp copy at  $x_0$  is a witness: lamps at distinct sites commute, and the compressed copy fixes the site. The theorem below then identifies the lamps along the whole  $L$ -orbit of  $x_0$  in every corona representation.

**Theorem S11.13** (involutive collapse). *Let  $H$  be countable, let  $L \leq H$  be a subgroup with property (T), and let  $s \in H$  satisfy  $s\gamma s^{-1} \in L$  for every  $\gamma \in L$ . Then*

$$D_{\text{coll}}(L, s) \leq \text{Rad}_{\text{MF}}(H).$$

*Proof sketch.* Fix a witness  $k$  and suppose, toward a contradiction, that some corona representation does not kill  $[\gamma_0, k]$  for some  $\gamma_0 \in L$ . Coordinate unitary lifts (Lemma 2.2) give an operator-norm almost representation  $(U_{g,n})$  on blocks  $M_{d_n}(\mathbb{C})$ , and after passing to a subsequence there is  $\eta > 0$  with  $\|U_{[\gamma_0, k], n} - 1\| \geq \eta$  for every  $n$ .

*Step 1: exact commuting involution microstates.* For  $\gamma \in L$  put  $v_\gamma = \gamma k \gamma^{-1}$ ; by (W1), (W3), and asymptotic multiplicativity their images asymptotically commute and square to 1. The spectral sign of the Hermitian part of a unitary  $u$  is an exact self-adjoint involution at operator distance at most  $\|u^2 - 1\|$  from  $u$ , and block compression along the finite commutative corner structure of the previously corrected microstates preserves exact commutation; induction over an enumeration of  $L$  yields exact pairwise-commuting self-adjoint involutions  $V_n(\gamma)$  with  $\|V_n(\gamma) - U_{v_\gamma, n}\| \rightarrow 0$  and  $\|U_{g,n} V_n(\gamma) U_{g,n}^* - V_n(g\gamma)\| \rightarrow 0$  for each fixed  $g \in L$ , where  $g\gamma$  denotes the orbit index of  $gv_\gamma g^{-1}$ .

*Step 2: the rank displacement metric.* For commuting exact involutions  $u, w$  the self-adjoint displacement  $T = u - w$  satisfies  $T^3 = 4T$ , so  $T^2/4$  is an orthogonal projection and  $\text{Tr}|u - w|^2 = 4 \text{rank}((u - w)^2/4)$ . Hence  $d_n(\gamma, \mu) = \text{rank}(V_n(\gamma) - V_n(\mu))$  is an integer-valued pseudometric; projections at operator distance below 1 have equal rank, so the covariance of Step 1 makes it eventually translation invariant:  $d_n(g\gamma, g\mu) = d_n(\gamma, \mu)$  for all large  $n$ , for each fixed triple. The rank comparison is applied to the projections  $T^2/4$ , not to the involutions themselves, and that is where the threshold is consumed: the displacements have norm at most 2, so a covariance error below  $1/8$  moves the associated projections by at most  $1/4 < 1$ .

*Step 3: generator normalization.* Property (T) makes  $L$  finitely generated; fix a Kazhdan pair  $(S, \kappa)$  with  $S$  finite symmetric and put  $k_n = \sum_{a \in S} d_n(1, a)$ . Translation invariance and the triangle inequality give  $d_n(1, \gamma) \leq |\gamma|_S k_n$  eventually, for the word length  $|\cdot|_S$ . If  $k_n = 0$  along a subsequence, the whole orbit collapses there to a single exact involution and  $\|U_{[\gamma_0, k], n} - 1\| \rightarrow 0$ , against the separation; so  $k_n \geq 1$  for all large  $n$ .

*Step 4: the displacement almost-cocycle.* In the Euclidean space of matrices set  $b_n(\gamma) = k_n^{-1/2}(V_n(\gamma) - V_n(1))$ . By Steps 2 and 3,  $\|b_n(\gamma)\|_{\mathbb{F}}^2 \leq 4|\gamma|_S$  at every large stage, with total generator mass

$$(11) \quad \sum_{a \in S} \|b_n(a)\|_{\mathbb{F}}^2 = 4$$

whenever  $k_n \geq 1$ . Conjugation by  $U_{a,n}$  is a Frobenius isometry. Fix  $a, \gamma \in L$ , put  $\mu = a^{-1}\gamma$ , and let  $k_n^{-1/2}D_n$  be the defect in the cocycle identity, so that

$$D_n = (V_n(\gamma) - V_n(1)) - U_{a,n}(V_n(\mu) - V_n(1))U_{a,n}^* - (V_n(a) - V_n(1)).$$

Two groupings of this one expression control the two factors of the Hilbert–Schmidt estimate, and neither yields the other. As written,  $D_n$  is a sum of three displacements of the commuting family of Step 1, one of them conjugated; rank is subadditive and invariant under unitary conjugation, so Step 2 with the bound of Step 3 gives

$$\text{rank } D_n \leq d_n(\gamma, 1) + d_n(\mu, 1) + d_n(a, 1) \leq (|\gamma|_S + |\mu|_S + |a|_S)k_n$$

eventually. Regrouped as

$$D_n = (V_n(\gamma) - U_{a,n}V_n(\mu)U_{a,n}^*) + (U_{a,n}V_n(1)U_{a,n}^* - V_n(a)),$$

the same  $D_n$  is a sum of two covariance errors of Step 1, so  $\|D_n\| \rightarrow 0$ . The rank bound comes from the first grouping and the operator-norm bound from the second. The norm bound alone would not give the rank bound: two exact involutions at small operator distance can differ in full rank. Feeding both into  $\text{Tr } |D|^2 \leq \text{rank}(D) \cdot \|D\|^2$  bounds the normalized defect mass by  $(|\gamma|_S + |\mu|_S + |a|_S) \|D_n\|^2 \rightarrow 0$ : the defect mass vanishes at the scale  $k_n$ . All estimates below concern only differences of the involutions.

*Step 5: bounded profile and an approximate primitive.* The Gaussians  $e^{-t\|\cdot\|^2}$  of the almost-cocycle are almost invariant under the twisted action, and the spectral gap of  $(S, \kappa)$  bounds the limiting profile  $\gamma \mapsto \lim_{n \rightarrow \omega} \|b_n(\gamma)\|_{\mathbb{F}}^2$ , the limit along a fixed free ultrafilter  $\omega$  on  $\mathbb{N}$ , by its generator values (11) — Delorme’s argument at the level of limiting seminorms. The limit must be taken along an ultrafilter and not as a limsup: positive definiteness of the Gaussians is transferred from the finite stages by evaluating a finite quadratic form with *signed* coefficients, which only a linear limit functional respects, and limsup gives the inequality in the wrong direction. The same linearity is what makes (11) an identity in the limit rather than an inequality. Since  $L$  is countable one could instead pass to a subsequence along which all the limits exist. A circumcenter

of the displacement orbit, taken at the level of bounded sequences, then produces one bounded sequence  $(w_n)$  with  $b_n(a) \approx w_n - U_{a,n}w_nU_{a,n}^*$  in limiting normalized mass, simultaneously for any prescribed finite set of movers.

*Step 6: transport through the compressor.* By (W2) the microstates of  $sas^{-1}$  and of 1 converge to one another, which is an operator-norm statement; Step 4 has just warned that two exact involutions at small operator distance can nevertheless differ in full rank, so this does not yet bound the displacement. What closes the gap is a rigidity property of the commuting family of Step 1: two *commuting* exact self-adjoint involutions  $u, w$  at operator distance  $\|u - w\| < 2$  are equal. Indeed  $T = u - w$  is self-adjoint with  $T^3 = 4T$ , so  $T^2/4$  is an orthogonal projection, whose norm is therefore 0 or 1; but  $\|T^2/4\| = \|T\|^2/4 < 1$ , so  $T^2 = 0$  and hence  $T = 0$ . The displacement therefore vanishes exactly on compressed elements, so including the compressed generators  $sas^{-1}$  among the movers makes the primitive almost fixed by them; hence  $u_n = U_{s,n}^*w_nU_{s,n}$  is almost fixed by every  $a \in S$ , because microstates of  $as^{-1}$  and  $s^{-1}(sas^{-1})$  agree asymptotically. Transporting forward as in Theorem 3.1 — compression to the almost-fixed spectral subspace of the averaged adjoint action of  $S$ , then the one-sided estimate  $P \leq VPV^*$  at the rescaled mass — returns  $w_n$  up to vanishing mass, again almost fixed by every  $a \in S$ . Then  $b_n(a) \approx w_n - U_{a,n}w_nU_{a,n}^*$  has small mass for every  $a \in S$  at a common large stage, contradicting (11). Normality of corona kernels extends the conclusion from the commutators to all of  $D_{\text{coll}}(L, s)$ .  $\square$

*Remark S11.14* (the quantitative form). As with the transport theorem, the limits in the sketch above can be replaced by explicit tolerances, the circumcenter by an approximate circumcenter of a near-optimal covering sequence, and the final contradiction assembled at one sufficiently late coordinate; we do not print that bookkeeping here. The commutation hypothesis (W3) is essential: it supplies the joint spectral calculus behind the rank identity of Step 2. Anticommuting Clifford coordinates have no joint spectral labels; their collapse mechanism is the central sign of Theorem A. At the corona level, Theorem S11.15 below removes the restriction to involutions, and Theorem S11.16 carries the collapse to witnesses of every finite order through their discrete Fourier idempotents.

The witness of Theorem S11.13 is a group element. The same collapse holds for arbitrary projections of the corona.

**Theorem S11.15** (projection collapse). *Let  $H$  be countable, let  $L \leq H$  have property (T), let  $s \in H$  satisfy  $s\gamma s^{-1} \in L$  for every  $\gamma \in L$ , and let  $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$  be a corona representation. Let  $p \in \mathcal{Q}$  be a projection such that  $p$  commutes with  $\Theta(s\gamma s^{-1})$  for every  $\gamma \in L$  and the projections  $\Theta(\gamma)p\Theta(\gamma)^*$ ,  $\gamma \in L$ , commute pairwise. Then  $p$  commutes with  $\Theta(L)$ .*

*Proof.* The sign  $w = 1 - 2p$  is a self-adjoint unitary of  $\mathcal{Q}$ . The subgroup  $H' \leq \mathcal{U}(\mathcal{Q})$  generated by  $\Theta(H)$  and  $w$  is countable. Inside  $H'$ , the image  $\Theta(L)$  has property (T),  $\Theta(s)$  compresses it,  $w$  is an involution centralizing the compressed copy, and the  $\Theta(L)$ -conjugates of  $w$  commute because the conjugated projections do. So  $w$  is an involutive compression witness in  $H'$ , and Theorem S11.13 places every commutator  $[\Theta(\gamma), w]$ ,  $\gamma \in L$ , in  $\text{Rad}_{\text{MF}}(H')$ . The inclusion  $H' \rightarrow \mathcal{U}(\mathcal{Q})$  is itself a corona representation of  $H'$ , so it maps each of these commutators to the identity:  $[\Theta(\gamma), w] = 1$  in  $\mathcal{Q}$ , and  $w = 1 - 2p$  gives  $[\Theta(\gamma), p] = 0$ .  $\square$

The witness here need not come from a group element: the projection is an arbitrary element of the corona. The relative commutant  $\Theta(sLs^{-1})' \cap \mathcal{Q}$  therefore cannot exceed  $\Theta(L)' \cap \mathcal{Q}$  by a projection with commuting  $L$ -orbit.

Projections in hand, functional calculus removes the involutivity of the witness altogether.

**Theorem S11.16** (torsion collapse). *Let  $H$  be countable, let  $L \leq H$  have property (T), let  $s \in H$  satisfy  $s\gamma s^{-1} \in L$  for every  $\gamma \in L$ , and let  $k \in H$  have finite order. If the compressed copy  $sLs^{-1}$  centralizes  $k$  and the conjugates  $\gamma k \gamma^{-1}$ ,  $\gamma \in L$ , commute pairwise, then every commutator  $[\gamma, k]$ ,  $\gamma \in L$ , is invisible in every corona representation of  $H$ ; the normal closure of all such commutators, over all finite-order witnesses for  $(L, s)$ , lies in  $\text{Rad}_{\text{MF}}(H)$ , with equality whenever the corresponding quotient of  $H$  is MF.*

*Proof.* Let  $\Theta$  be a corona representation,  $m$  the order of  $k$ , and  $u = \Theta(k)$ , a unitary with  $u^m = 1$ . Its discrete Fourier idempotents  $p_j = m^{-1} \sum_{t \in \mathbf{Z}/m} \zeta^{-jt} u^t$ , for  $\zeta = e^{2\pi i/m}$  and  $j \in \mathbf{Z}/m$ , are self-adjoint idempotents of the corona. Each  $p_j$  is a polynomial in  $u$ , so whatever commutes with  $u$  commutes with every  $p_j$ : the represented compressed copy centralizes each  $p_j$ , and the  $\Theta(L)$ -conjugates of each  $p_j$  commute pairwise because the conjugated witnesses do. Theorem S11.15 therefore makes every  $p_j$  commute with  $\Theta(L)$ , and Fourier reconstruction expresses  $u$  as a  $\zeta$ -weighted sum of the  $p_j$ , so  $u$  commutes with  $\Theta(L)$  and  $\Theta([\gamma, k]) = 1$  for every  $\gamma \in L$ .  $\square$

The involutive collapse has a second structural consequence: killing the central sign of the Kazhdan–Clifford group of Theorem A makes all lamps commute, and the moved root lamp becomes an involutive witness outright.

**Theorem S11.17** (the commuting-lamp quotient). *The quotient  $W/\langle \zeta \rangle$  of the Kazhdan–Clifford group by its central sign is sofic and is not MF. More precisely, the image of the commutator  $[v_1, tct^{-1}]$ , the inverse of the marked commutator  $u$ , in  $W/\langle \zeta \rangle$  is nontrivial and lies in  $\text{Rad}_{\text{MF}}(W/\langle \zeta \rangle)$ ; that last conclusion holds with  $W$  replaced by the ambient affine–Clifford group of any injective endomorphism  $\alpha$  of a countable property-(T) group and any base element outside the image of  $\alpha$ .*

*Proof.* Write  $q: W \rightarrow W/\langle \zeta \rangle$  and  $k = q(tct^{-1})$ , the moved root lamp. Since  $\zeta$  is killed, any two lamps commute in the quotient, so the  $q(L)$ -conjugation



orbit of  $k$  commutes pairwise, where  $L = q(\bar{\Gamma})$  is the image of the base copy;  $k$  remains an involution, and the compressed copy  $q(t)Lq(t)^{-1}$  centralizes  $k$  because the compressed base fixes the moved site. Property (T) passes to the image  $L$ , so Theorem S11.13 places every commutator  $[q(\bar{v}_1), k]$  in the MF radical of the quotient. Nontriviality survives the sign quotient: in the signed finite-support model of the lamp kernel, the commutator is the product of the two distinct lamp basis elements at  $\tau o$  and  $j(\bar{v}_1)\tau o$ , whose configuration coordinate is nonzero, while every element of  $\langle \zeta \rangle$  has configuration coordinate zero.

Killing  $\zeta$  abelianizes the lamp kernel:  $\text{Cilamp}(X)/\langle \zeta \rangle$  is the elementary abelian group  $\bigoplus_X \mathbb{Z}/2$  under the same  $V$ -action, so  $W/\langle \zeta \rangle \cong (\bigoplus_X \mathbb{Z}/2) \rtimes V$ . That kernel is again locally finite with finite level orbits, so the argument of Theorem D gives soficity unchanged.  $\square$

Every route to an MF model of  $W$  through the commuting-lamp quotient is thereby closed, and  $W/\langle \zeta \rangle$  is a second finitely generated sofic group that is not MF. The same collapse reaches the finitely presented group.

*Proof of Theorem B.* Write  $q: E \rightarrow E/\langle w \rangle$ ,  $\bar{d} = q(d)$ , and  $\bar{L} = q(\iota(\mathcal{B}))$ . The element  $\bar{d}$  is an involution because  $c^2 = 1$ , and  $q(t)\bar{L}q(t)^{-1}$  centralizes  $\bar{d}$  because  $t\iota(\mathcal{B})t^{-1}$  commutes with  $c$  in  $E$ . Property (T) passes to  $\bar{L}$ , and  $q(t)$  compresses it. Conditions (W1) and (W2) of Definition S11.12 therefore hold, and the content is (W3).

The rotation letters commute with  $d$ , and  $t$  carries each  $v_i$  to  $v_i^2$ , so  $\iota(v_i^2)$  commutes with  $d$  as well. Hence  $\iota(g)d\iota(g)^{-1}$  depends only on the exponent vector of the translation part of  $g$  modulo 2, which leaves eight conjugates, indexed by  $(\mathbb{Z}/2)^3$ . By Theorem 5.1 the marked word is  $w = [d, \iota(v_1)d\iota(v_1)^{-1}]$ , the commutator attached to the class of  $v_1$ . The six rotation words  $x, x^2, z, yz, yx, y^2x$  carry that class onto the six remaining nonzero classes, and each commutes with  $d$ ; conjugating the displayed identity by the corresponding  $\iota(r)$  exhibits every nonzero parity commutator as a conjugate of  $w$ , while the zero class contributes  $[d, d] = 1$ . Centrality of  $w$  puts all eight in  $\langle w \rangle$ , and one further conjugation reduces an arbitrary pair of conjugates to this case. Thus  $\bar{d}$  is an involutive compression witness for  $(\bar{L}, q(t))$ , and Theorem S11.13 places  $[q(\iota(v_1)), \bar{d}]$  in  $\text{Rad}_{\text{MF}}(E/\langle w \rangle)$ .

Nontriviality passes through the homomorphism  $E \rightarrow W$  of Proposition S3.3: it carries  $[\iota(v_1), d]$  to the product of the lamp basis elements at the two distinct sites  $\tau o$  and  $j(\bar{v}_1)\tau o$ , and carries  $\langle w \rangle$  into  $\langle \zeta \rangle$ , whose configuration coordinate is zero. Hence  $[\iota(v_1), d] \notin \langle w \rangle$ , its image in  $E/\langle w \rangle$  is nontrivial, and Proposition S11.3 makes  $E/\langle w \rangle$  non-MF. Corollary S11.5 identifies  $\text{Rad}_{\text{MF}}(E)$  with the  $q$ -preimage of  $\text{Rad}_{\text{MF}}(E/\langle w \rangle)$ , so  $u$  belongs to it.  $\square$

The Clifford sign is thus a certificate for an obstruction that survives its removal, and  $E/\langle w \rangle$  is a second finitely presented non-MF group.



**Corollary S11.18** (collapse computes the radical). *In the setting of Theorem S11.13, write  $q: H \rightarrow H/D_{\text{coll}}(L, s)$  for the quotient map. Then, unconditionally,*

$$\text{Rad}_{\text{MF}}(H) = q^{-1}(\text{Rad}_{\text{MF}}(H/D_{\text{coll}}(L, s))),$$

*and if  $H/D_{\text{coll}}(L, s)$  is MF, then  $\text{Rad}_{\text{MF}}(H) = D_{\text{coll}}(L, s)$ .*

*Proof.* Combine Theorem S11.13 with Corollaries S11.5 and S11.4.  $\square$

In the second case  $H/D_{\text{coll}}(L, s)$  is the universal MF quotient of  $H$  (Proposition S11.3): the corona representation theory of  $H$  is computed exactly by the quotient.

**Theorem S11.19** (defect saturation). *In the setting of Theorem S11.13:*

- (1) *Every homomorphism  $f$  from  $H$  to a countable MF group maps every element of  $D_{\text{coll}}(L, s)$  to the identity.*
- (2) *If  $D_{\text{coll}}(L, s) = H$ , then  $\text{Rad}_{\text{MF}}(H) = H$ ; every homomorphism from  $H$  to a countable MF group is trivial; and if  $H$  is nontrivial, then  $H$  is not MF.*

*Proof.* (1) By Theorem S11.13 and Lemma S11.2,

$$f(D_{\text{coll}}(L, s)) \leq f(\text{Rad}_{\text{MF}}(H)) \leq \text{Rad}_{\text{MF}}(f(H)).$$

The subgroup  $f(H)$  of the MF target is MF (Lemma S11.9(1)), so its radical is trivial by Proposition S11.3. (2) The radical statement is Theorem S11.13 with  $D_{\text{coll}}(L, s) = H$ ; triviality of every homomorphism follows from (1); and a nontrivial group with full radical is not MF by Proposition S11.3.  $\square$

A group as in (2) admits no nontrivial finite-dimensional unitary representation, no nontrivial MF quotient, and no nontrivial homomorphism to any countable residually finite or amenable group. The strategy — detect a defect analytically, then saturate it algebraically — develops a suggestion of Fournier-Facio (personal communication, 2026; cf. the Questions preamble of Section S12). Whether a small-cancellation construction produces a finitely presented group with saturated collapse or normal-Kazhdan defect is open.

**S11.5 The reduced group algebra of  $E$**  Theorem C was stated in Section 7; its stable finiteness comes from the following standard facts, recorded here because the faithfulness of the canonical trace is what carries the argument.

**Lemma S11.20.** *Let  $G_1$  be a countable discrete group.*

- (1) *The canonical tracial state  $\tau$  on  $C_{\text{red}}^*(G_1)$  is faithful.*
- (2) *Every unital  $C^*$ -algebra  $A$  with a faithful tracial state is stably finite: for every nonempty finite index set  $I$ , every isometry in  $M_I(A)$  is a unitary. In particular this holds in  $M_k(A)$  for every  $k \geq 1$ .*

*Proof.* (1) Realize  $C_{\text{red}}^*(G_1) \subseteq B(\ell^2 G_1)$  as the norm closure of the span of the left translation unitaries  $\lambda_g$ . Let  $\rho_g$  be the right translation unitaries,  $(\rho_g \xi)(h) = \xi(hg)$ . Each  $\rho_g$  commutes with each  $\lambda_{g'}$ , hence with every element of  $C_{\text{red}}^*(G_1)$ . The canonical trace is  $\tau(x) = \langle x\delta_e, \delta_e \rangle$ . If  $\tau(x^*x) = 0$  then  $\|x\delta_e\| = 0$ , so  $x\delta_e = 0$ , and then  $x\delta_g = x\rho_{g^{-1}}\delta_e = \rho_{g^{-1}}x\delta_e = 0$  for every  $g \in G_1$ . The vectors  $\delta_g$  are total in  $\ell^2 G_1$ , so  $x = 0$ .

(2) On  $M_k(A)$  the functional  $\tau_k([x_{ij}]) = \frac{1}{k} \sum_i \tau(x_{ii})$  is a state with the trace property  $\tau_k(xy) = \frac{1}{k} \sum_{i,j} \tau(x_{ij}y_{ji}) = \tau_k(yx)$ , and it is faithful: the  $i$ -th diagonal entry of  $x^*x$  is  $\sum_j x_{ji}^*x_{ji}$ , so  $\tau_k(x^*x) = 0$  forces  $\tau(x_{ji}^*x_{ji}) = 0$ , hence  $x_{ji} = 0$ , for all  $i, j$ . If  $v \in M_k(A)$  satisfies  $v^*v = 1$ , then  $y := 1 - vv^*$  is a projection with  $\tau_k(y) = \tau_k(1) - \tau_k(vv^*) = \tau_k(1) - \tau_k(v^*v) = 0$ ; since  $y = y^*y$ , faithfulness gives  $y = 0$ , i.e.  $vv^* = 1$ .  $\square$

**S11.6 A sofic non-MF group** The same quotient that proves  $w \neq 1$  also provides the sofic non-MF group of Theorem D.

*Proof of Theorem D.* Write  $T = \varinjlim (\bar{\Gamma}, \bar{\alpha})$ , so  $V = T \rtimes \mathbb{Z}$ , and let  $B \leq V$  be the level-zero copy of  $\bar{\Gamma}$ . The image of  $\bar{\alpha}$  has finite index (at most eight, by the parity classes of the translation lattice). Thus every telescope level  $\bar{\Gamma}_n \leq T$  is commensurable with  $B$ . The stable letter conjugates  $B$  onto  $\bar{\alpha}(\bar{\Gamma})$ , so all of  $V$  commensurates  $B$ . Orbit-stabilizer now shows that every  $\bar{\Gamma}_n$ -orbit in  $X = V/B$  is finite.

Consider a finite subset of  $\text{CLamp}(X) \rtimes T$ . Its  $T$ -coordinates lie in one level  $\bar{\Gamma}_n$ . Each of its lamp coordinates has finite  $\bar{\Gamma}_n$ -orbit: this holds for the central sign and for each site lamp, hence for every Clifford word. The finitely many resulting orbits generate a finite  $\bar{\Gamma}_n$ -invariant subgroup  $L \leq \text{CLamp}(X)$ , because  $\text{CLamp}(X)$  is locally finite. Our original finite set therefore lies in  $L \rtimes \bar{\Gamma}_n$ , which is sofic (finite-by-sofic). Soficity is local over finite subsets, so  $\text{CLamp}(X) \rtimes T$  is sofic.

Reassociating the semidirect products gives

$$W = \text{CLamp}(X) \rtimes (T \rtimes \mathbb{Z}) \cong (\text{CLamp}(X) \rtimes T) \rtimes \mathbb{Z}.$$

The final group is sofic because a split extension of a sofic group by  $\mathbb{Z}$  is sofic. Truncate the integer levels to  $\mathbb{Z}/N$  and let the shift act cyclically. Induce the kernel's sofic approximations over this finite level set. The cocycle identity that makes the finite-quotient case exact now fails only on the wrap-around boundary. For a window shifted by  $k$ , that boundary occupies at most  $2|k|/N$  of the levels, which vanishes as  $N$  grows. This is the  $\mathbb{Z}$ -extension case of the Elek–Szabó amenable-extension theorem [ES06]. Here  $\bar{\Gamma}$  is sofic: its six generators lie in  $\text{GL}_4(\mathbb{Z})$ , the resulting integral subgroup is residually finite through its congruence quotients, and it is isomorphic to  $\bar{\Gamma}$ .

The map  $E \rightarrow W$  of Proposition S3.3 is surjective: its image contains  $V$ , the root lamp  $c_o$ , and hence every lamp by transitivity of the  $V$ -action on  $X$ . Since  $E$  is generated by  $v_1, x, y, z, t, c$  (by the Tietze calculation above), their six images generate  $W$ . The map sends  $w \in \text{Rad}_{\text{MF}}(E)$  to the nontrivial

sign  $\zeta$ ; functoriality (Lemma S11.2) makes  $\zeta \in \text{Rad}_{\text{MF}}(W)$ . Hence  $W$  is not MF.  $\square$

Failure of extension permanence. MF groups are not closed under extensions. More precisely,  $W$  fits into an exact sequence

$$1 \longrightarrow \text{ClLamp}(X) \longrightarrow W \longrightarrow V \longrightarrow 1$$

whose kernel and quotient are MF, while  $W$  is not. The concrete lamp kernel is locally finite, LEF, sofic, and MF, while the total witness is finitely generated and non-MF.

*Proof.* By Lemma S11.9(3), the Clifford lamp group is locally finite, hence MF. The group  $V$  identifies with the subgroup of  $\text{GL}_4(\mathbb{Q})$  generated by  $\bar{\Gamma}$  and  $D$ : the telescope levels are  $D^{-n}\bar{\Gamma}D^n$ , and the determinant gives the  $\mathbb{Z}$ -coordinate. It is residually finite. At every odd modulus the determinant 8 of the dilation is invertible, so the congruence reduction of the integral affine model descends to conjugation by the dilation and extends over the telescope and its shift. These reductions separate any nontrivial element; compare Mal'cev's theorem [Mal] for finitely generated linear groups. Thus  $V$  is MF by Lemma S11.9(2), whereas  $W$  is not MF by the preceding construction.  $\square$

Extensions by  $\mathbb{Z}$ . Composing the two canonical projections gives a surjection  $W \rightarrow \mathbb{Z}$  whose kernel is  $\text{ClLamp}(X) \rtimes T$ , and this kernel is LEF, hence MF: every finite subset lies in  $L \rtimes \bar{\Gamma}_n$  for a finite invariant lamp subgroup  $L$  and one telescope level, and that group is residually finite because the level acts on the finite group  $L$  through a finite quotient. Thus  $W$  is an extension of an LEF group by  $\mathbb{Z}$  and is not MF, while soficity is preserved by every extension with amenable quotient [ES06].

An exact stably finite non-MF algebra. The same witness gives a concrete separable exact example:  $C_{\text{red}}^*(W)$  is exact, has a faithful tracial state, is stably finite, and is not MF. Indeed, the locally finite lamp kernel is amenable and hence exact, while the linear group  $V \leq \text{GL}_4(\mathbb{Q})$  is exact [GHW]. Exact groups are closed under extensions [KW2], so  $W$  is exact; for discrete groups this is equivalent to exactness of the reduced group algebra [KW1]. The canonical trace and stable finiteness follow as in Lemma S11.20. Finally, an MF embedding of  $C_{\text{red}}^*(W)$  would restrict to a faithful norm-corona model of the canonical group unitaries, contradicting that  $W$  is not MF.

A simple sofic envelope. There is a countable simple sofic group  $S_{\text{simp}}$  with

$$\text{Rad}_{\text{MF}}(S_{\text{simp}}) = S_{\text{simp}}.$$

Equivalently, every homomorphism from  $S_{\text{simp}}$  to an MF group is trivial.

*Proof.* By Elek–Szabó, every countable sofic group embeds in a countable simple sofic group [ES05, Theorem 1]. Embed  $W$  in such a group. Functoriality of the MF radical sends  $\zeta$  to a nontrivial element of the MF radical of the simple group. That radical is normal, so it is the whole group. Every homomorphism to an MF group kills the MF radical.  $\square$

**S11.7 Marked-limit closure** The explicit clopen cylinder constructed in the next subsection is an instance of a general phenomenon: at every fixed number of marked generators, the MF property is closed under limits of marked groups, and its failure always has a finite certificate.

**Theorem S11.21** (the MF locus is closed). *Fix  $k \geq 1$  and consider the space of  $k$ -marked groups — quotients of the free group  $F_k$ , topologized so that a basic clopen set fixes the truth of finitely many equations among words in the generators.*

- (1) *The marked groups with MF quotient form a closed set.*
- (2) *Equivalently, the non-MF marked groups form an open set.*
- (3) *Every non-MF  $k$ -marked group admits a finite-radius certificate: there is a radius  $R$  such that every  $k$ -marked group whose relations agree with it on the reduced words of length at most  $R$  is non-MF.*

*Proof.* A failure of MF for the quotient is a failure of normalized local approximation, and such a failure is witnessed on one finite subset of the quotient. Expressing the finitely many multiplications and inequalities of that subset by words in the marked generators produces finitely many relation constraints; every marked group agreeing on those constraints transports the failed local-model problem verbatim, so an entire basic clopen neighborhood fails MF. Enclosing the constraint words in a reduced-word ball gives the radius form.  $\square$

Thus failure of MF is always certified by one finite ball of the marked Cayley diagram. The next subsection computes such a certificate explicitly for  $E$ .

**S11.8 A finite universal Horn obstruction** Theorem A can be read as one first-order fact about all MF groups at once: a finite quasi-identity, that is, a single universal implication between finitely many equations (a universal Horn sentence). Write  $R$  for the forty-one relators of (8), regarded as words in eight variables, and  $w$  for the marked word in those variables.

**Proposition S11.22** (finite quasi-identity). *Every MF group  $G$  satisfies the finite universal Horn sentence*

$$\forall x_1, \dots, x_8: \bigwedge_{r \in R} r(x_1, \dots, x_8) = 1 \implies w(x_1, \dots, x_8) = 1,$$

*while the canonical generating tuple of  $E$  satisfies every relator in  $R$  and has  $w \neq 1$ .*

*Proof.* A tuple satisfying the relators induces a homomorphism  $E \rightarrow G$  sending the marked word of  $E$  to  $w(x_1, \dots, x_8)$ . If  $G$  is MF, that image is trivial:  $w \in \text{Rad}_{\text{MF}}(E)$  by Theorem A, and the radical is functorial (Lemma S11.2). The canonical tuple of  $E$  satisfies  $R$  by definition, and its marked word is nontrivial by Theorem A(1).  $\square$

**Corollary S11.23** (a basic clopen set of non-MF marked groups). *In the space of eight-generator marked groups — quotients of the free group  $F_8$ , topologized so that a basic clopen set fixes the truth of finitely many equations among words in the generators — the marked groups satisfying all relators of  $R$  and  $w \neq 1$  form a nonempty clopen set consisting entirely of non-MF groups.*

*Proof.* Both conditions constrain finitely many values of the marked multiplication, so the set is clopen; it contains the marked group  $E$  by Theorem A(1), and Proposition S11.22 excludes every MF point.  $\square$

Thus the obstruction of Theorem A is finite, local, and locally constant on a basic clopen neighborhood in the marked-group topology.

**S11.9 Undecidability of MF recognition** Lemma S11.9(1) and Theorem A make MF a Markov property of finitely presented groups: the trivial group is MF, and by subgroup heredity the finitely presented group  $E$  embeds in no MF group. The Adian–Rabin theorem [Ra] therefore applies.

**Corollary S11.24** (undecidability). *There is no algorithm that decides, from a finite presentation, whether the presented group is MF. By the effective form of the Adian–Rabin construction [Ra], the set of finite presentations of non-MF groups is not even recursively enumerable.*

The Adian–Rabin construction [Ra] is quoted here and used nowhere else. It supplies a computable map taking an instance  $(P, w)$  of the word problem of a fixed finitely presented group with undecidable word problem to a finite presentation whose group is trivial — hence MF — when  $w = 1$ , and contains  $E$  — hence non-MF, by subgroup heredity — when  $w \neq 1$ . The instances with  $w = 1$  form a recursively enumerable set which, the word problem being undecidable, is not recursive; hence the instances with  $w \neq 1$  form a set that is *not* recursively enumerable. An enumeration of the non-MF presentations would enumerate that set, making the word problem decidable. No completeness of the word problem under many-one reduction is needed, only its undecidability; this is the pullback below. Everything group-theoretic that feeds the construction is machine-checked — subgroup heredity of MF and the finitely presented non-MF group — and so is the computability-theoretic pullback: from any computable Adian–Rabin reduction, undecidability and the failure of recursive enumerability on the negative side follow formally. Theorem S11.21 certifies single non-MF groups by finite balls; the corollary says no algorithm classifies the positive and negative cases from presentations.

The phenomenon also has infinite multiplicity, in two senses that need no new analysis. The groups  $E \times \mathbb{Z}^k$  for  $k \geq 0$  are finitely presented, pairwise nonisomorphic — their abelianizations have distinct torsion-free ranks, equivalently distinct numbers of homomorphisms to  $\mathbb{Z}/2$  — and non-MF, each containing  $E$  (Lemma S11.9(1)). Beyond finite presentation, pairing  $E$  with the continuum of pairwise nonisomorphic two-generator groups of

B. H. Neumann [Ne] yields  $2^{\aleph_0}$  pairwise nonisomorphic finitely generated non-MF groups: each product  $E \times N$  is non-MF, and a fixed countable group  $X$  satisfies  $X \cong E \times N$  for at most countably many of the continuum many isomorphism types  $N$  — each such  $N$  embeds in  $X$  as a finitely generated subgroup, a countable group has only countably many finitely generated subgroups, and the types  $N$  are pairwise nonisomorphic. A map with countable fibres from a continuum-sized family has continuum-sized image. The counting step is needed because the products can fail to be pairwise nonisomorphic even when the factors are. Neumann's family is quoted, not reproved [Ne].

## Part IV. Limits of the method, and further apparatus

### S12 LIMITATIONS OF THE OPERATOR-NORM METHOD

The proof is specific to the operator norm. If multiplication is controlled only in normalized Hilbert–Schmidt norm, the induced adjoint operators on the Hilbert–Schmidt spaces need not be close in operator norm. Both transport proofs require exactly that operator-norm control: the ultraproduct proof needs it to obtain an honest representation on the ultraproduct, and the finite-stage proof of Section S5 needs it for the spectral estimates. Normalized Hilbert–Schmidt control therefore does not supply the transport used here. The failure is exact and dimension-free: against the operator-norm estimate  $\|\mathrm{Ad}(U) - \mathrm{Ad}(V)\| \leq 2\|U - V\|$ , there are unitaries of arbitrarily small normalized Hilbert–Schmidt distance whose adjoints stay 2 apart in operator norm — the same constant, in the opposite direction.

The converse dilution is exact. Padding a model with an identity block preserves every operator-norm quantity — the multiplicative defects and the pairwise separations are unchanged — while the normalized Hilbert–Schmidt distances decay with the padding ratio: below any prescribed bound, every operator-norm local model can be replaced by one all of whose pairwise Hilbert–Schmidt distances are smaller than the bound. Hilbert–Schmidt invisibility of an element alone therefore can never prove a group non-MF: an obstruction must produce an invariant substructure carrying its own scale, renormalized intrinsically so that the normalization survives dilution, rather than reading a normalized trace off the ambient blocks. Macroscopic relative rank is *not* what is required, and none of the constructions here has it: padding leaves the Reynolds corner  $q_n M_{d_n}(\mathbb{C}) q_n$  untouched while  $d_n$  grows, so its relative rank  $r_n/d_n$  may tend to zero, and the argument runs on the corner’s own normalized trace. The moving spectral subspace and the rank-calibrated collapse renormalize on their own range and on the displacement scale  $k_n$  in the same way.

This is a limitation of the method, not a tracial nonapproximation theorem for  $E$ . Alekseev–Thom ask whether commutants arising in sofic approximations of Kazhdan groups can be recovered from finite-dimensional coordinate subalgebras [AT, Open Problem 6.2]. Such a coordinate description is separate from the operator-norm theorem proved here.

Finite dimensionality of the coordinates is essential in a second, sharper sense: it cannot be traded for stable finiteness of the coordinate algebras together with faithful traces. In a general faithfully traced unital  $C^*$ -algebra the transport argument would act on the trace Hilbert space by the left and right regular actions, and the von Neumann algebra these two actions generate can fail every finiteness property — for an infinite-dimensional  $\mathrm{II}_1$  factor in standard form it is all bounded operators on the trace space. Constant coordinates  $A_n = C_{\mathrm{red}}^*(E)$  refute any such generalization outright:



the left regular representation embeds  $E$  exactly and injectively into the corresponding corona, the element  $c$  centralizes the Kazhdan base there, and conjugation by  $t$  produces  $tct^{-1}$ , whose commutator with the base is precisely the nontrivial defect that Theorem A detects. A faithfully-traced coordinate transport theorem would collapse that commutator; no such theorem can hold. The refuted statement is the transport theorem of Section 3 with its matrix coordinates replaced by finite faithfully traced  $C^*$ -coordinates and its normalized Hilbert–Schmidt norm by the trace 2-norm, every other hypothesis copied verbatim.

**Questions** The finite-normal criterion has a precise torsion limitation: every finite subgroup of a torsion-free group is trivial. By Theorem S7.6, a normal property-(T) subgroup of the defect subgroup obstructs corona representations without any torsion. What remains for a torsion-free example is group-theoretic: a torsion-free finitely presented group whose defect subgroup contains a normal subgroup with property (T). Fournier-Facio, who removed the analogous torsion obstacle for nonsoficity [FFF], has suggested (personal communication, 2026; cf. Remark 5.2) that a small-cancellation construction should produce such a group.

- (1) What is  $\text{Rad}_{\text{MF}}(E)$ ? It properly contains  $\{1, w\}$  by Theorem B. Applying Corollary S11.18 to  $E/\langle w \rangle$  with the witness of that proof reduces the computation to the quotient by the resulting collapse defect  $D$ , and by Corollary S11.4 identifies the radical with the preimage of  $D$  as soon as that quotient is MF. Whether it is MF is open; a detecting model must survive both the sign collapse of Theorem A and the commuting-orbit collapse, and by Theorem S8.1 it factors through no finite quotient and no exact finite-dimensional representation.
- (2) Is there a torsion-free finitely presented non-MF group? The analytic obstruction no longer requires torsion (Theorem S7.6); the remaining input is group-theoretic. A construction along the lines suggested above — strengthening Hull’s common-quotient technique so that the target elements are chosen inside a given infinite normal subgroup of a torsion-free acylindrically hyperbolic group, applied to Fournier-Facio’s skeleton — would yield a torsion-free finitely presented group equal to its own MF radical, so that every homomorphism to every norm matrix corona is trivial. The implication from the construction’s data to the conclusion is proved: given such routing data, there is a two-generated, finitely presented, torsion-free property-(T) group equal to its own MF radical, and every nontrivial quotient of such a group also fails to be MF. Both statements assume the routing data; neither is unconditional. Constructing the routing data uses the Hull–Osin small-cancellation theory and remains open; no unconditional torsion-free finitely presented non-MF group is proved here. By Theorem S11.19, a group whose collapse or normal-Kazhdan defect saturates admits no nontrivial homomorphism to any countable MF group at all.

- (3) Is  $E$  sofic, or hyperlinear? The adjoint spectral estimates require operator-norm multiplicative control, and neither property is decided by the results above.
- (4) Does MF imply hyperlinearity? A faithful embedding into a norm matrix corona need not make any coordinate trace faithful, so the defining MF representation does not immediately supply a hyperlinear model. This question concerns the CDE norm-corona convention fixed above; for stronger trace-controlled MF conventions the implication is built into the definition.
- (5) By Proposition S11.7, every family of theorems placing subgroups in the MF radical generates, by iteration through quotients, a closure contained in  $\text{Rad}_{\text{MF}}$ . For which groups do the rules of Theorems S7.3, S7.6, S11.13, and S11.15 compute the radical exactly?

### S13 THE MAXIMAL GROUP $C^*$ -ALGEBRA

The familiar construction of the maximal group  $C^*$ -algebra completes the group ring over all unitary representations at once, and “all” is a class rather than a set. This appendix carries out the standard size-honest version, relative to a fixed set-theoretic universe  $\mathfrak{U}$ ; the result is the usual maximal group  $C^*$ -algebra. For a  $\mathfrak{U}$ -small group  $H$ , define  $C_{\max, \mathfrak{U}}^*(H)$  to be the closed  $*$ -subalgebra generated by the diagonal copy of  $H$  in the bounded product of all its nontrivial unitary representations on  $\mathfrak{U}$ -small unital  $C^*$ -algebras. We suppress the subscript  $\mathfrak{U}$  and write  $u_h$  for the canonical unitary associated with  $h \in H$ .

**Proposition S13.1** (universe-relative maximal group algebra). *The canonical map  $h \mapsto u_h$  is injective. For every  $\mathfrak{U}$ -small unital  $C^*$ -algebra  $B$  and every homomorphism  $\rho: H \rightarrow \mathcal{U}(B)$ , there is a unique unital  $*$ -homomorphism  $C_{\max, \mathfrak{U}}^*(H) \rightarrow B$  mapping  $u_h$  to  $\rho(h)$  for every  $h \in H$ .*

For a nontrivial target, projection to its coordinate gives the extension; for a trivial target, the map is unique. Density of the generated  $*$ -subalgebra gives uniqueness, while the left-regular coordinate proves injectivity.

The maximal and reduced algebras separate already at the level of finiteness, by an algebraic mechanism:

**Proposition S13.2** (proper compression forbids finiteness). *Let  $A$  be a unital  $C^*$ -algebra (a unital star ring suffices) containing a projection  $p$  and a unitary  $u$  such that the conjugate  $upu^*$  absorbs  $p$  on both sides and  $upu^* \neq p$ . Then  $A$  contains an isometry that is not a unitary; consequently  $A$  is not stably finite and carries no faithful tracial state.*

*Remark S13.3* (the maximal algebra under strict Kazhdan compression). For a group carrying a strict Kazhdan compression  $t\Gamma t^{-1} \subsetneq \Gamma$  — the base of  $E$  is the running example — the Kazhdan projection  $p_\Gamma$  constructed in Section 3 exists in the maximal algebra, conjugation by  $u_t$  dominates it, and the domination is strict: at the quasi-regular coordinate  $\ell^2(G/t\Gamma t^{-1})$  the point mass at the base coset is fixed by the compressed subgroup and

moved by every element outside it, so the two projections differ there. By Proposition S13.2 the maximal algebra of such a group is then not stably finite, carries no faithful trace, and in particular is neither residually finite-dimensional nor MF. The assembly runs through the spectral projection, in the maximal algebra, of the averaged generator built from the Kazhdan pair: the spectral gap places the projection, the quasi-regular coordinate separates it from its conjugate, and Proposition S13.2 applies verbatim. The contrast with Theorem C is the point of this remark: for the reduced algebra of  $E$  the canonical trace is faithful and the algebra is stably finite, so its failure of MF is not a failure of finiteness; for the maximal algebra, infiniteness is forced by the strict compression before any MF considerations arise.

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